



Computational Reproducibility Review for Community Conferences with CODECHECK

IPIN 2026

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2026-10-06 | <https://t1p.de/ipin-check>

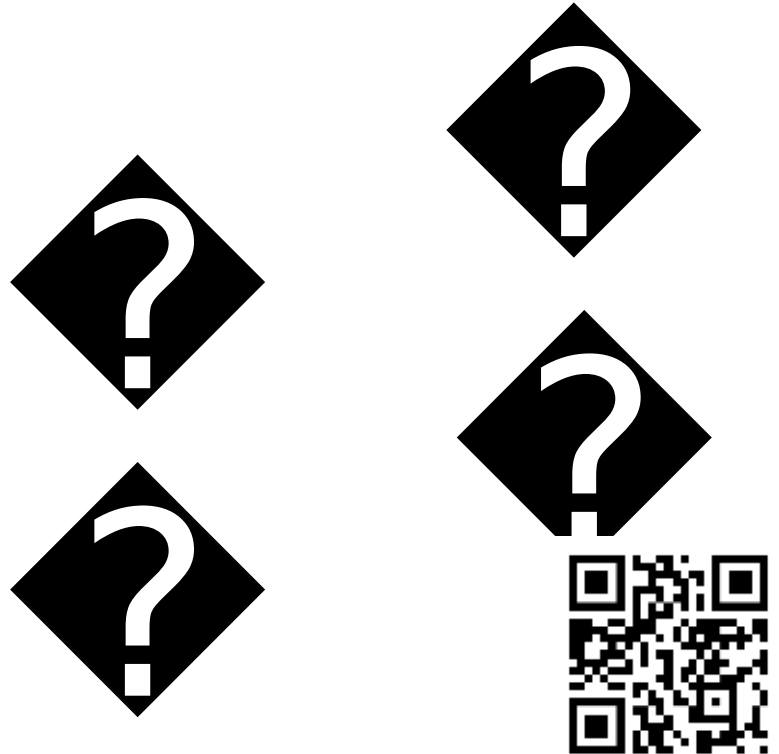


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Closed and irreproducible research



Claerbout's claim:

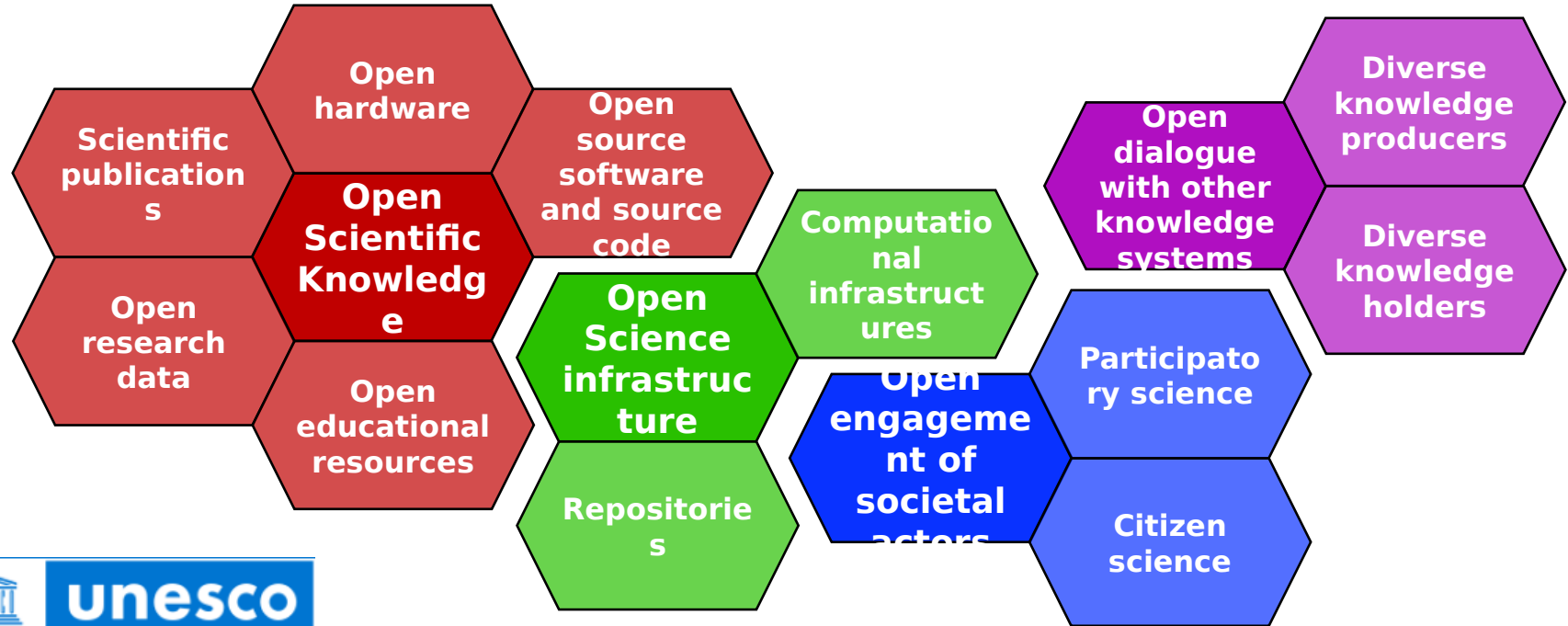
An article about computational science in a scientific publication is not the scholarship itself, it is merely advertising of the scholarship. The actual scholarship is the complete software development environment and the complete set of instructions which generated the figures.

<https://doi.org/10.1190/1.1822162>

https://doi.org/10.1007/978-1-4612-2544-7_5



What are Open Science components?



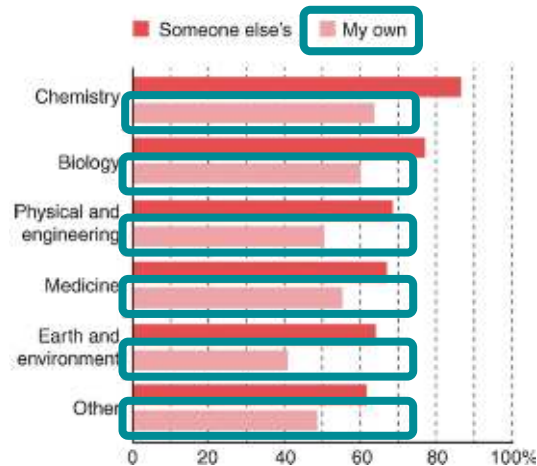
unesco

Further reading:

[UNESCO Recommendation on Open Science](#)

A reproducibility crisis?

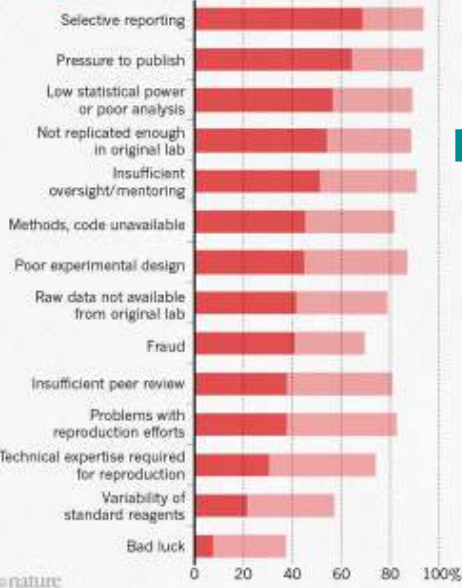
Most scientists have experienced failure to reproduce results



WHAT FACTORS CONTRIBUTE TO IRREPRODUCIBLE RESEARCH?

Many top-rated factors relate to intense competition and time pressure.

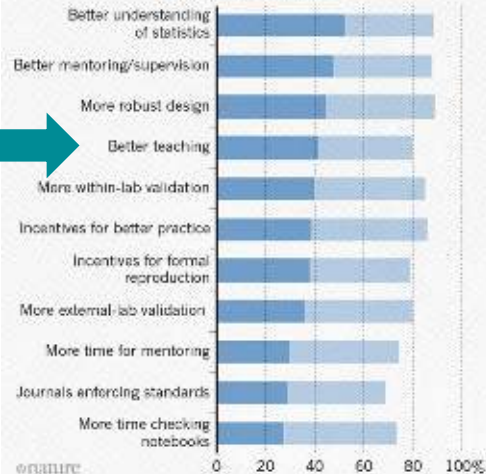
Legend: ■ Always/often contribute ■ Sometimes contribute



WHAT FACTORS COULD BOOST REPRODUCIBILITY?

Respondents were positive about most proposed improvements but emphasized training in particular.

Legend: ■ Very likely ■ Likely



Much ado about reproducibility?

[PLoS Med.](#) 2005 Aug; 2(8): e124.

Published online 2005 Aug 30. doi: [10.1371/journal.pmed.0020124](https://doi.org/10.1371/journal.pmed.0020124)

Why Most Published Research Findings Are False

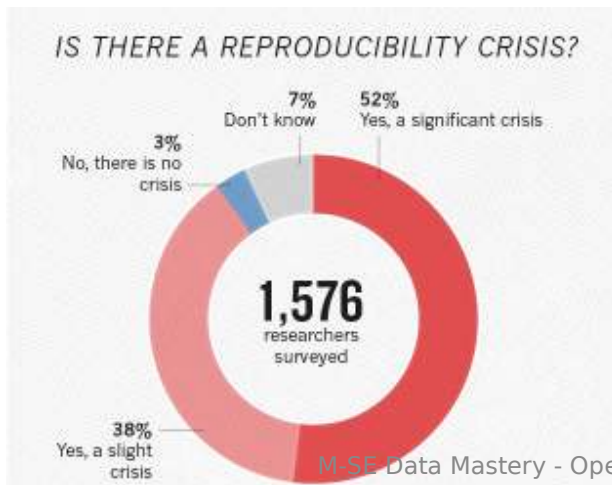
[John P. A. Ioannidis](#)

Published: 25 May 2016

1,500 scientists lift the lid on reproducibility

Monya Baker

Nature **533**, 452–454 (2016) |



Science has been in a “replication crisis” for a decade. Have we learned anything?

Bad papers are still published. But some other things might be getting better.

By [Sally Gunter](#) | Oct. 14, 2020, 12:20pm EDT

<https://www.vox.com/future-perfect/21504366/science-replication-crisis-peer-review-statistics>

Matters Arising | Published: 14 October 2020

Transparency and reproducibility in artificial intelligence

Benjamin Haibe-Kains , George Alexandru Adam, Ahmed Hosny, Farnoosh Khodakarami, Massive Analysis Quality Control (MAQC) Society Board of Directors, Levi Waldron, Bo Wang, Chris McIntosh, Anna Goldenberg, Anshul Kundaje, Casey S. Greene, Tamara Broderick, Michael M. Hoffman, Jeffrey T. Leek, Keegan Korthauer, Wolfgang Huber, Alvis Brazma, Joelle Pineau, Robert Tibshirani, Trevor Hastie, John P. A. Ioannidis, John Quackenbush & Hugo J. W. L. Aerts

Nature **586**, E14–E16 (2020) | [Cite this article](#)

by **Will Douglas Heaven**

November 12, 2020

<https://www.technologyreview.com/2020/11/12/1011944/artificial-intelligence-replication-crisis-science-big-tech-google-deepmind-facebook-openai/>



How to draw an owl

1.



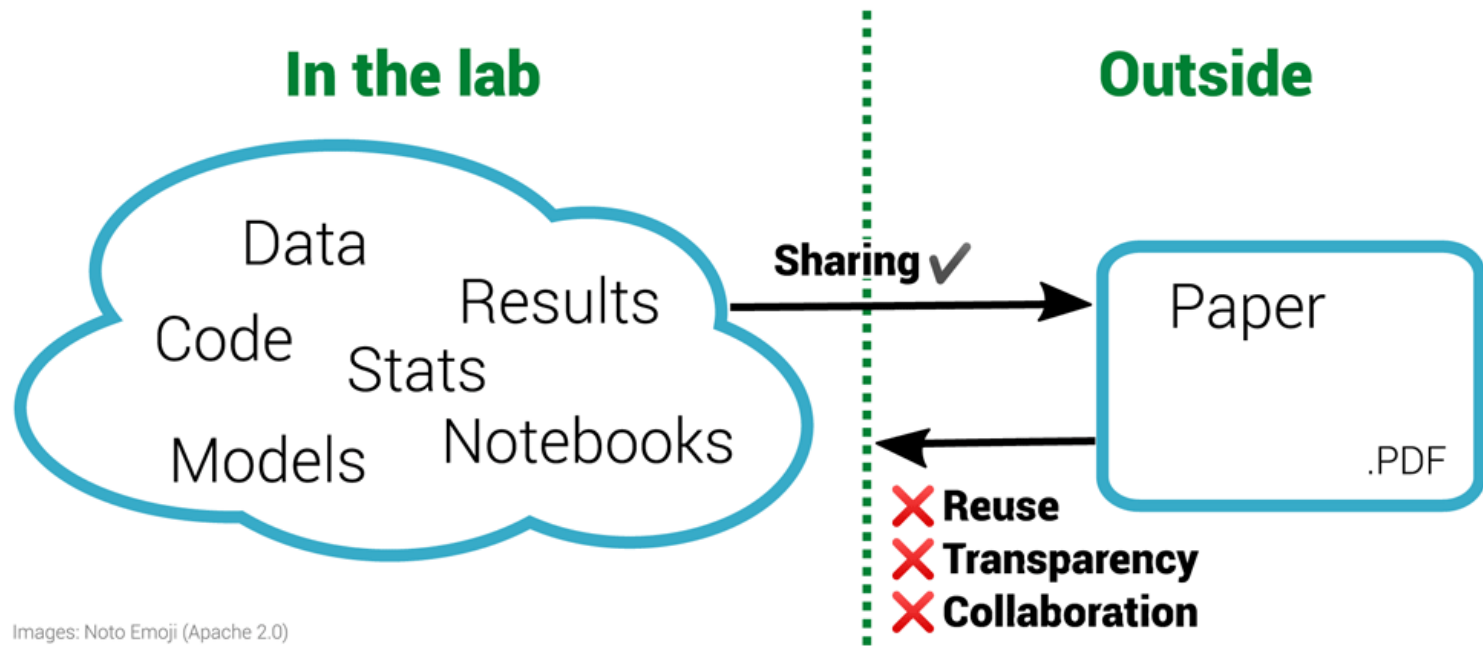
1. Draw some circles

2.



2. Draw the rest of the fucking owl





The inverse problem in reproducible research. Figure 1 of <https://doi.org/10.12688/f1000research.51738.1>

The left half of the diagram shows a diverse range of materials used within a laboratory. These materials are often then condensed for sharing with the outside world via the research paper, a static PDF document. Working backwards from the PDF to the underlying materials is impossible. This prohibits reuse and is not only non-transparent for a specific paper but is also ineffective for science as a whole. By sharing the materials on the left, others outside the lab can enhance this work.

One thing

about reproducible research

Have a README: all else is details.

Show willingness to help, but don't stop publishing because of lacking docs, your code is **good enough**. It is hard to document for someone else, therefore, document for future you, add more **details on demand**.

“Rule” inspired by Greg Wilson’s Teaching Tech Together (<http://teachtogether.tech/en/index.html>), Rule 1.

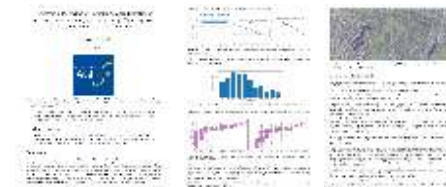
One re-execution of computational workflow by codechecker during scholarly communication, review & publication



Independent execution of computations underlying research articles.



1. Codecheckers record but don't investigate or fix.
2. Communication between humans is key.
3. Credit is given to codecheckers.
4. Workflows must be auditable.
5. Open by default and transitional by disposition.



130+ Certificates

<https://codecheck.org.uk/register/>



CODECHECK Certificate 2022-018

10.5281/zenodo.7084333

Rainer Sika

September 27, 2022



Table 1: CODECHECK summary

This certificate was generated by the CODECHECK system, which is a part of the R package *codecheck*. It provides a summary of the integration of the CODECHECK system into the R package *codecheck*. The integration was performed by the R package *codecheck* and the R package *codecheck* is available on CRAN.

Authors: Rainer Sika, David Ganev, Jatin Bala, Andrew T. Papadopoulos
 Reference: <https://doi.org/10.5281/zenodo.7084333>
 Summary: The CODECHECK system provides a summary of the integration of the CODECHECK system into the R package *codecheck*. The integration was performed by the R package *codecheck* and the R package *codecheck* is available on CRAN.

Table 2: Summary of output files generated

File	Description
Figure 1b.pdf	Figure 1b of the article
Figure 1b.pdf	Figure 1b of the article
Figure 1b.pdf	Figure 1b of the article
Figure 1b.pdf	Figure 1b of the article
Figure 1b.pdf	Figure 1b of the article

Summary

The summary of the figure, from the R package *codecheck*, is reproduced below.

CODECHECK notes

Data and Code

The repository was not possible to be accessed. The repository was not possible to be accessed. The repository was not possible to be accessed.

Software Installation

The provided code was not possible to be executed. The provided code was not possible to be executed. The provided code was not possible to be executed.

Running the Script

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References

Figure 1b: David Ganev, Jatin Bala, and Andrew T. Papadopoulos. 2022. "Data and Codes for the Manuscript of the R package *codecheck*." Zenodo. <https://doi.org/10.5281/zenodo.7084333>

Colophon

This document was built with Quarto.

Session Info

codecheck

Version: 0.0.1 (2022-09-27)

Platform: x86_64-pc-linux-gnu (64-bit)

Running under: Ubuntu 22.04.1 LTS

Matrix product: default

RAN: /usr/lib/x86_64-linux-gnu/libc.so.2.34.0

LAKE: /usr/lib/x86_64-linux-gnu/libc.so.2.34.0

Locale:

[1] LC_CTYPE=en_US.UTF-8 LC_MESSAGES=C
 [2] LC_TIME=en_US.UTF-8 LC_MESSAGES=en_US.UTF-8
 [3] LC_NUMERIC=en_US.UTF-8 LC_MESSAGES=en_US.UTF-8
 [4] LC_COLLATE=en_US.UTF-8 LC_MESSAGES=C
 [5] LC_MONETARY=en_US.UTF-8 LC_MESSAGES=C
 [6] LC_TELEPHONE=en_US.UTF-8 LC_MESSAGES=C
 [7] LC_MEASUREMENT=en_US.UTF-8 LC_MESSAGES=C

attached base packages:

[1] stats graphics gtable MASS datasets methods base

loaded via a namespace (and not attached):

[1] digest 0.0.28 jsonlite 1.8.0 magrittr 2.0.3 rmarkdown 2.16
 [2] httr 1.4.2 stringr 1.4.1 dplyr 1.0.7
 [3] R6 2.5.1 Rcpp 1.0.10 compiler 4.2.1
 [4] RcppEigen 0.12.97

Figure 1b: Data and Codes

Figure 1b: Data and Codes

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Figure 1b: Data and Codes

Figure 1b: Data and Codes

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Figure 1b: Data and Codes

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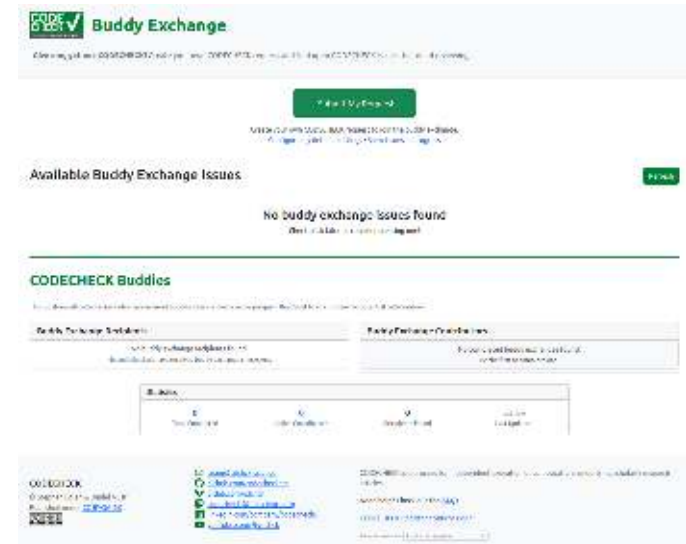
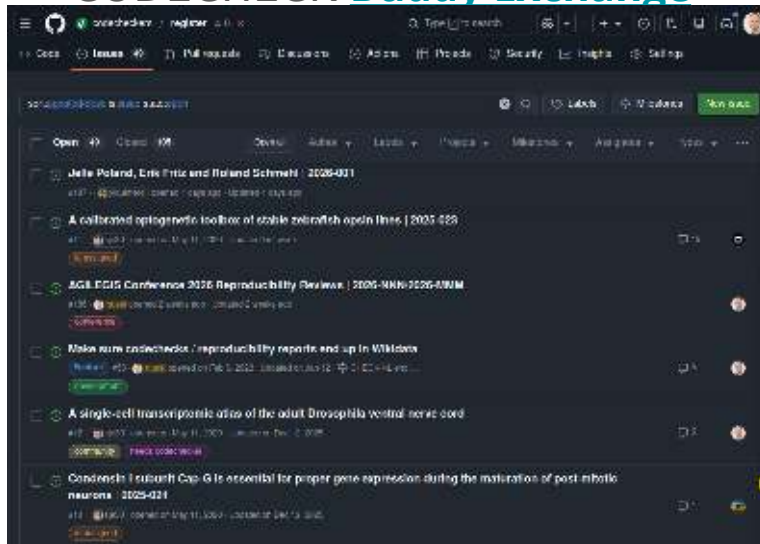
Figure 1b: Data and Codes

Figure 1b: Data and Codes

Figure 1b: Data and Codes

Now how do I get my certificate?

1. Submit to a **venue** that regularly conducts CODECHECKs > AGILE GIS conference
2. Loop us into a conversation with an **editor** to any journal/event
3. Open an issue in the [register repository](#) to get a **community check**, or better...
4. CODECHECK **Buddy Exchange**



CODECHECK Community Workflow

<https://codecheck.org.uk/guide/community-workflow-overview>

For authors

Have a README, find a suitable checklist to use, basic metadata in codecheck.yml, licensing information; acknowledgement in manuscript

For codecheckers (short, extended, in-person events)

Handling repos/files, approaching a workspace/workflow, responsibilities, writing the certificate;

it is a **craft**, so **we mentor** better than we write process descriptions!

For editors (me)

CODECHECK Register

[illegible]

Your work here

<https://codecheck.org.uk/register/>

Currently at CODECHECK

AGILE Reproducibility Committee 2026 completed

New conversations and community collaborations (IPIN, criminology)

CHECK-PUB completed - <https://codecheck.org.uk/pub/>

Open Journal Systems (OJS) Plugin to streamline CODECHECK integration:

collect metadata, create register entries, display certificate info;

[Community CODECHECKs](#) & [volunteer list](#)

Updated Jupyter certificate template, CODECHECKs in Zotero and Wikidata, register improvements, ...

CHECK-PUB / ojs-codecheck

CODECHECK Demo Journal

Home » Journals » Vol. 1 No. 2020 Issue 1 » Codecheck

The principal components of natural Images

Author: Eva Nuhn
Author: Franziska König
Author: Sabine Timpf

Published: 2020-10-26

Issue: Vol. 1 No. 2 / 2020 Issue 2

Section: Articles

Abstract

Most navigation systems for pedestrians output the shortest route. However, there are findings that travellers do not use the shortest route when free to choose. One alternative to minimising spatial distance is the incorporation of landmarks information in a shortest route algorithm. Yet, we do not know whether pedestrians prefer such a landmark route over the shortest route. Therefore, we perform a survey and show participants videos of a shortest and a landmark route. We let participants answer questions concerning navigation satisfaction, route communication, and route comparison. Our findings show that the landmark route is more favourable.

Data and Software Availability

Platform & availability by OJS / PKP

Open Journal Systems (OJS) | <https://github.com/codecheckers/ojs-codecheck/>
<https://codecheck.org.uk/pub/>

[illegible]

<https://codecheck.wikibase.cloud/>

[illegible]



73 certificates = AGILE
reproducibility review reports

Historic coincident community creation

> time to merge processes and branding?



[Home](#)
[Check](#)
[Vendors](#)
[People](#)
[Organizations](#)
[Statistics](#)
[About](#)

[CODECHECK Register](#)
[Vendors](#)
[Conferences](#)
[AGILE Conference of Geographic Information Science](#)

CODECHECKs for AGILEGIS

Website: <https://agile-gis.eu>

Website: [conference](#)

API statistics: [index.html](#)

Content: [AGILE GIS pages](#)



AGILE is the Association of Geographic Information Laboratories in Europe, a membership organization connecting geographic information researchers and educators across the continent. It runs an annual international conference on geographic information science, alongside PhD schools, research initiatives and open science programmes. AGILE places particular emphasis on reproducible research practice in GIScience.

Certificate	Report	Work	Check date
2020-0108	doi.org/10.7705/OSF.IO/2TC1M	Integrating collision avoidance and dynamic global grid systems: a case study into vehicle modelling	2020-07-13
2020-0118	doi.org/10.7705/OSF.IO/5SVMT	What is do in the Moonline: A Service Coverage Analysis for Parked Autonomous Vehicles	2020-07-13
2020-020	doi.org/10.7705/OSF.IO/71WH2	Window Operators for Processing Spatial-Temporal Data Streams on Unmanned Vehicles	2020-07-13
2020-021	doi.org/10.7705/OSF.IO/SUWVJ	Comparing supervised learning algorithms for Spatial Normal Entity recognition	2020-07-13
2020-022	doi.org/10.7705/OSF.IO/XR0G	Extracting interrogative intents and concepts from geo-analytic questions	2020-07-13
2020-023	doi.org/10.7705/OSF.IO/XD5VH	Tracking Hurricane Dorian in GDFIT and Twitter	2020-07-13
2021-002	doi.org/10.7705/OSF.IO/42Rm	Building Change Detection of Airborne Laser-Scanning and Dense Image Matching Point Clouds using Height and Class Information	2021-06-10
2021-003	doi.org/10.7705/OSF.IO/d4S2a	Investigating drivers' perceptual abilities in urban environments	2021-06-10
2021-004	doi.org/10.7705/OSF.IO/42Rm	Extraction of linear structures from data: linear models using deep learning	2021-06-10

Who is AGILE?

<https://agile-online.org/>

Association of Geographic Information Laboratories
in Europe (<https://agile-online.org/>)

Annual AGILE conference (

<https://agile-online.org/conference-2020>)

Bi-annual PhD School (

<https://agile-online.org/agile-actions/phd-school>)

AGILE Initiatives (

<https://agile-online.org/funding-initiatives>)

Collaboration & MoU with organizations & sister
associations (

<https://agile-online.org/agile-community/cooperation>

)

AGILE Association of Geographic Information Laboratories in Europe

HOME AGILE COMMUNITY CONFERENCE 2023 PAST CONFERENCES & PROCEEDINGS AGILE ACTIVITIES FUNDING INITIATIVES FAQ

Members

Search For: (e.g. City, Organisation, Person) In AGILE membership

Name	City	Country
S2 North	Münster	Germany
Orb Lab/Arch Geographic Data Institute, School of Geography & Sustainable Development, University of St Andrews	St Andrews, Scotland	United Kingdom
Bonn University of Applied Sciences, Department of Geodesy	Bonn	Germany
Leiden University Institute	Vlaardingen	Netherlands
Catholic University of Leuven	Leuven	Belgium
Centro AGILEM, Faculdade de Engenharia, Universidade do Porto	Porto	Portugal



<https://reproducible-agile.github.io/>

2017, '18 & '19: Workshops on reproducibility

2019: Reproducible publications at AGILE conferences
(AGILE Initiative supported by AGILE Council)

2020: 1st AGILE reproducibility review (guidelines optional)

2021: 2nd AGILE reproducibility review (guidelines mandatory)

2022: 3rd AGILE reproducibility review + Replication Workshop

...

2026: 7th AGILE reproducibility review (desk reject w/o DASA section)

Review process



Proceedings: agile-giscience-series.net/review_process.html

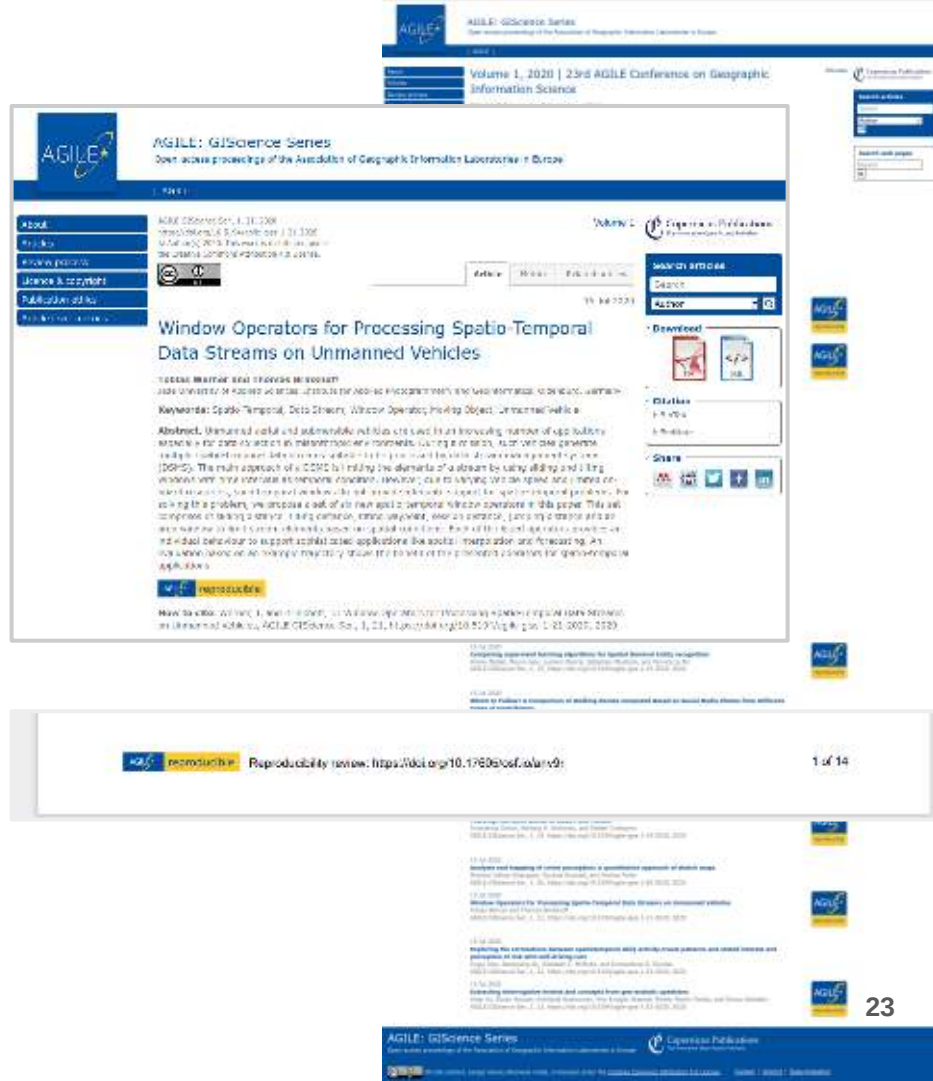
Process documentation: osf.io/7rjpe/

Reproducibility review of **full papers** *after* accept/reject decisions by scientific reviewers

Reproducibility review & communication

Community conference & volunteers

Badges on proceedings website and article title page link to **reproducibility reports** (Thanks Copernicus!)



AGILE Reproducible Paper Guidelines

(v1)

<https://doi.org/10.17605/OSF.IO/CB7Z8>

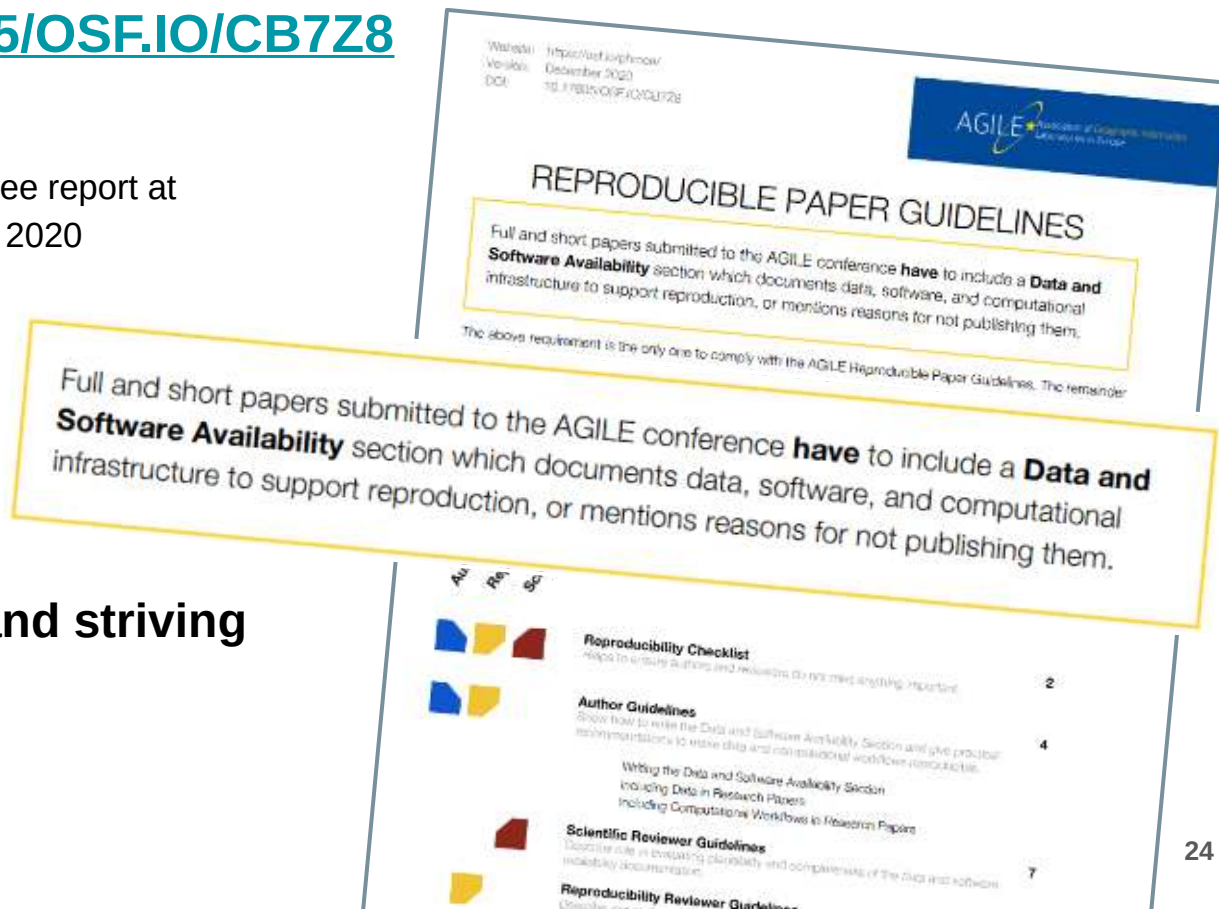
Created by AGILE Initiative in 2019 (see report at <https://osf.io/hupxrl>) and updated in 2020

Transparency over
enforcement

Promote, don't exclude

Acknowledge spectrum and striving
for ideal

Tailored to GIScience



The guidelines

<https://doi.org/10.17605/OSF.IO/CB7Z8>

Reproducibility checklist

Author guidelines

Writing DASA section

Data in Research Papers

Computational workflows in Research Papers

Reviewer guidelines

Reproducibility reviewer guidelines

Background

REPRODUCIBLE PAPER GUIDELINES

Full and short papers submitted to the AGILE conference **have** to include a **Data and Software Availability** section which documents data, software, and computational infrastructure to support reproduction, or mentions reasons for not publishing them.

The above requirement is the only one to comply with the AGILE Reproducible Paper Guidelines. The remainder of the document provides concrete recommendations for all involved stakeholders to increase transparency, reproducibility, and openness of Computational GIScience research. The following table of contents shows the recommended parts for different readers. Familiarity with all sections is, of course, beneficial.

Author	Reproducibility Reviewer	Scientific Reviewer
		
Reproducibility Checklist		
Helps to write authors and reviewers do not miss anything important.		
2		
		
Author Guidelines		
What to do to write the Data and Software Availability Section, and give practical recommendations to make data and computational workflows reproducible.		
Writing the Data and Software Availability Section Including Data in Research Papers Including Computational Workflows in Research Papers		
		
Scientific Reviewer Guidelines		
Recommendations on reviewing reproducibility and data and software availability documentation.		
		
Reproducibility Reviewer Guidelines		
Describe the review approach to assess workflows and their effects.		
Background		
10		

Further resources

These guidelines can not cover all details of the reproducibility review at AGILE conferences. For more information for authors, translators, and practical examples see the [guidelines.pdf](https://osf.io/cb7z8/). For more information about the review process and deadlines, see the [process description](https://osf.io/cb7z8/). For any questions, please visit the AGILE Reproducible Reviewers' forum for the [Reproducible Paper Guidelines](https://osf.io/cb7z8/).

Checklist and writing the DASA section

REPRODUCIBILITY CHECKLIST

For all **datasets** included/produced in the paper, check if data:

- ☐ Is provided in a non-proprietary format.
- ☐ Is documented for third parties to reuse.
- ☐ Is accessible in a public repository and has an open data license.

For all **software tools/libraries/packages** and **computational workflows** included/produced, check if:

- ☐ Reproduction steps are explained in a README (plain text file), flowchart, or script.
- ☐ Computational environments (including hardware) are documented or provided.
- ☐ Versions of relevant software components (libraries, packages) are provided.
- ☐ All parameters and expected execution times for the computational workflow are provided.
- ☐ Software developed by the authors is available in a public repository and has an open license.
- ☐ There is a clear connection between **tables, figures, maps, and statistical values** and the data and code that they are based on, e.g., using file names or documentation in the README.

In the **Data and Software Availability** section, check if you include:

- ☐ Data and software statements (see examples below).
- ☐ The reasons, if any, for not being able to share (parts of) data or code.

For all **data and software** check that:

- ☐ All datasets and code (used or mentioned) are assigned DOIs.
- ☐ Datasets and code are cited throughout the paper.

After acceptance in the **camera-ready paper** check that:

- ☐ If data has been shared privately or anonymously for peer review, they are updated with all metadata and accessible via a DOI and referenced from the paper.
- ☐ If a reproducibility review report will be published for your paper, a DOI URL in the Data and Software Availability section is included using the following template:
A reproducibility report for this paper is available confirming that [consistently name of the computational workflow / its results / Figure 1 and 2] could be independently reproduced, see <https://doi.org/10.xxxxxx> for report.

WRITING THE DATA AND SOFTWARE AVAILABILITY SECTION

The DASA section provides references to where data, software and documentation is available (e.g., paper section or README file) and under what conditions (e.g., copyright, licenses or access procedures for protected data). It should be concise and contain persistent links to repositories using Digital Object Identifiers¹ (DOI). You may remove links for anonymity during peer review ("xxx"), or share anonymized links² if your repository supports them. Data, software and (third-party) tools should be cited following recommended citation or standard citation guidelines. Possible statements for the DASA section are provided below. You may include one of these statements or draft your own.

Statements for non-computational or conceptual work

No data or code was collected, developed, or used in this work.

The full list of reviewed literature is available at [link to attachment or citable deposit of bibliography].

The full concept maps are available at [link] and the ideas were first sketched in a blog post at [link].

Research data/code supporting this publication ...

... is available in [name of the repository(-ies)] and is accessible via the following DOI [DOI link(s)]

... was accessed on [date of dataset access/download] with the following [query parameters, if applicable] under the license [dataset license].

... was downloaded manually using the services at [name of organisation] (using a departmental subscription for costs) and [name of organisation]. The compiled dataset cannot be redistributed due to licensing restrictions.

... is not available due to [indicate reasons, e.g., licenses, sensitive data on human subjects, privacy statements]; if there are processes to obtain the data, describe them.

The computational workflow supporting this publication ...

... is executed via [choose, e.g., a single command/file, a workflow management software, a set of numbered scripts], published under license [the license] at [DOI of repository].

... is published in a [language] module/package at [link of software project]. The used version is archived at [DOI of repository].

... is provided as a [container/VM] published at [DOI of repository] with instructions included in the file README.md in the repository.

The guidelines for data



“What if...” and
Examples not shown

INCLUDING DATA IN RESEARCH PAPERS

	Minimum requirements	Recommended practices
What?	• All input data and configuration • Data description/documentation, including provenance, field or column types, etc. • If data is retrieved from an external source, documentation on collection queries and download steps	• Standardised, discipline-specific metadata⁸ and ontologies to describe your data • Data download scripts
Where?	• Publish data in a public repository providing a DOI • Cite data (including date and version) in the paper	• Discipline- or data type-specific repository⁹ • Include recommended citation in dataset description (unless already provided by repository) • Create a registration for OSF projects¹⁰ and use the DOI to cite it
How?	• Use open data formats; export from proprietary format for publication • Specify the license	• Use plain text-based file formats

The guidelines for computational workflows



INCLUDING COMPUTATIONAL WORKFLOWS IN RESEARCH PAPERS

	Minimum requirements	Recommended practices
What? Computational environment	- Describe the used environment and computational infrastructure, e.g., hardware specs, operating system - List software versions - Cite used software¹⁴	- Provide the actual environment, e.g., a Dockerfile + container¹⁵ or a Virtual Machine (e.g., using OSGeo-Live) - Provide a pinned freeze of your dependencies (structured configuration files with dependency information) - Add a colophon or "reproducibility receipt"¹⁶ to your notebooks - Installation and execution instructions for different operating systems
Computation steps	- Document the detailed steps in a text file and/or flowchart (every action/click) - Document expected execution times given computing power unless negligible - Ask a colleague to try out the instructions	- Scripts/models and a README file that explains their use - All figures are fully scripted and a peer has read your README's instructions (incl. interactive visualisations and interactive adjustments) - Multi-panel plots are composited with scripts¹⁷ - Software package with structured metadata¹⁸, tests/Ci¹⁹, and a pipeline framework²⁰ or workflow language²¹ - Live documents for analyses, e.g., Binder²² - Live demo of APIs/online applications (e.g., anonymous cloud resources, such as Google Cloud Run or AWS) - Subset or a synthetic dataset for quick evaluation
Where?	Repository providing a persistent identifier, e.g., a DOI or SWHID²³	Versioned code repository, such as GitHub or GitLab, and ongoing open development
How? Tools used	Use generally available tools (avoid proprietary tools that are not available to reviewers and other researchers)	- Use and create Open Source tools - Cite core modules/tools/language used
Development practices	- Use clear licenses²⁴ that fit your environment - Follow one of "Good enough practices in scientific computing"²⁵	Follow all "Good enough practices." Use development guidelines for your environment / language of choice (e.g., for R²⁶)

Scientific reviewer guidelines... concerning the reproducibility review only!



SCIENTIFIC REVIEWER GUIDELINES

This section clarifies the expectations and role of the scientific reviewer with respect to the reproducible paper guidelines. For information for the Reproducibility Reviewer, please see the following section.

Reproducibility is considered good scientific practice that provides input for the quality assessment of a paper. Therefore, reviewers of AGILE papers should be aware of the **author guidelines on reproducibility** and be familiar with the **reproducibility checklist**, as well as the expected content of the **mandatory data and software availability section**. Using this information, reviewers should evaluate the plausibility and completeness of the data and software availability documentation, and whenever possible and readily available **include feedback on reproducibility aspects** in their comments. Scientific reviewers are free to but **are not expected to attempt reproductions of computations**.

Data and software availability documentation provide an additional set of information for assessing the quality of research presented in a manuscript. Reviewers are asked to know about the AGILE reproducible paper guidelines and to consider the level of reproducibility reached in a manuscript. To do so, they shall assume the position of someone who would like to reproduce the submitted work to assess whether the provided material is likely to allow reproduction of the submitted work. Based on this impression, reviewers may challenge authors regarding the level of reproducibility reached, if any statements are made regarding reproducibility in a manuscript.

Scientific reviewers are not required to actually reproduce a manuscript, but, if the data and code are provided in an anonymous format, and if a reviewer attempts to reproduce all or parts of the submitted work, then they are asked to document the process and outcomes (see Reproducibility Reviewer Guidelines below). Please reach out to the reproducibility chair if you are keen on conducting a reproducibility review for a paper you are reviewing.

The peer review of AGILE papers is a fully anonymous peer review, i.e. authors and reviewers do not know each other's identity. Reviewers should be supportive to authors and consider potential limitations in access to resources due to anonymisation. Since the provision of information to help reproduction of a paper can accidentally lead to disclosure of an author's identity, the reviewers should not use any such additional information to the disadvantage of the authors. The reviewers' comments provided to the authors are expected to be neutral²⁶ and contribute to improved reproducibility of the reported findings.

The guidelines for reproducibility reviewers

Ideal vs. realistic

Role

Skills

Do's & don't's

REPRODUCIBILITY REVIEWER GUIDELINES

Reproducibility reviewers conduct a complimentary review of the workflow that is published with a manuscript. Ideally, reproducibility reviewers only **read the abstract and the Data and Software Availability section** (DASA) of an article. They may read other sections referenced in the latter. Then they follow the authors' instructions for executing the workflow, ideally starting from the DASA or a README file in the referenced reproduction material. When reproducibility reviewers get stuck, they take advantage of the option to **communicate** with the authors early and often. Reproducibility reviewers should be aware of the different levels for making research reproducible in the author guidelines (see above) to be able to **recommend improvements** to the author and at the same time have the skillset and tools to conduct their review efficiently. Reproducibility reviewers are not responsible for making a workflow transparent or executable. Reproducibility reviewers **write a short reproducibility report** documenting their communication and the results of their reproduction attempt. This report is published if the reproduction was, at least in part, successful.

The reproducibility review from a reproducibility reviewer's perspective

While these AGILE reproducible paper guidelines are created with an intention to eventually have 100% of computations of accepted submissions succeed, understanding, and ultimately community adoption, tasks harder and progress slower yet hopeful accepted article, but a successful reproduction should be aware of this role and accept that a reviewer and the scientific reviewer on the same

The scope of the reproducibility review is not community is worth taking a look at for further review. A partial reproduction, i.e. if you can see as a success at this point, though what fellow reproducibility editors or the reproducibility

Reproducibility reviewer skills

A reproducibility review ideally is a learning experience for the AGILE community to increase openness concrete amount of time you spend on a reproduction of research you are tasked to reproduce get things to work within minutes (no counting an hour to get a workflow started. Although a good enough for anyone to reproduce a workflow package managers and getting familiar with DESCRIPTION files and renv for R, npm for JavaScript

Do	Don't
Quick pre-pro-review checks and ask authors to fix before continuing, even if not all of these are technically required, authors who are willing to work reproducibly can show their engagement right from the start: 1. Do the links to data sets and materials resolve? 2. Is there a README with clear step-by-step instructions? 3. Is there a clear mention of to be expected execution times? 4. Is there a LICENSE file to ensure openness?	Dig across badly or un-documented collections of files and functions to identify which part of the code/data creates which figure/table/output; find or build the "start button" yourself.
Encourage authors by pointing out promising intermediate results or concrete benefits of reproducibility.	Run workflows requiring considerable computational resources (unless interesting for you) but ask for data subsets for demonstration purposes.
Accept sample datasets to run a workflow and compare the outcome with the expected sample results; check the sources of the full datasets, if available.	Accept private sharing of data or code, unless strictly required for protection of sensitive data. All changes by the author should update to the public reproduction material.
Clearly document the extent of the reproduction in your reproduction report and suggest potential improvements; if you provide intermediate feedback, to include a history of your interactions in the report so that the ideas you contributed are preserved when the submission's material is improved.	Attempt to install software without any instructions, install binary software of unknown origin, or try to fix installation problems you encounter on your machine; try to install without (a) asking for help from a fellow reproducibility reviewer who is familiar with the software, or (b) asking the author to help, providing a minimal reproducible example of your problem.
Get in touch with fellow reproducibility reviewers if specific expertise (tool, programming language, ...) is needed.	Point out or even fix problems that are not specific to the submission, e.g., general problems in a software tool.
Set an example when communicating about computational problems, e.g., by clearly defining your system (OS version, language version, etc.)	Create accounts on any service or platform to access code, data, or other resources.

Example reproducibility review reports from AGILE 2026



Reproducibility review of: Inclusive Multimodal Routing: Who Gets Left Behind?

Egor Kotov

2026-03-15



This report is part of the reproducibility review at the AGILE conference. For more information see <https://reproducibility.agile.ac.uk/>.

This document is published on OJS at <https://ojs.agile.ac.uk/117835/OJS-10642761>.

To cite the report see:

Kotov, E. (2026). Reproducibility review of Inclusive Multimodal Routing: How Biophysical Constraints Shape Accessibility. <https://doi.org/10.1101/2026.03.15.117835>

Reviewed paper:

Gargano, L., Gargano, M., Almaghi, N., and Gargano, L. Inclusive Multimodal Routing: How Biophysical Constraints Shape Accessibility. AGILE 2026, Vol. 2, <https://doi.org/10.1101/2026.03.15.117835>

Submission Title:

Inclusive Multimodal Routing: Who Gets Left Behind?

Post Review Title:

Inclusive Multimodal Routing: How Biophysical Constraints Shape Accessibility

Authors:

Gargano, L., Gargano, M., Almaghi, N., Gargano, L.

Ref. Paper:

<https://doi.org/10.1101/2026.03.15.117835>

3 Reproduction Efforts

During the reproduction study, the following issues were identified and resolved to achieve full reproduction:

3.1 1. Hardcoded CPU Allocation on Shared HPC Nodes

The original script explicitly hardcoded the multiprocessing pool size to `sp.cpu_count() + 1`. This caused HPC resource issues on shared nodes. The fixed number of logical cores on the physical node (e.g., 192) was used instead of the actual number of cores allocated to the job. This caused severe thread thrashing. The associated pipeline patched the script to respect the SLURM `CPUS` task resource allocation.

3.2 2. Extreme Computational Demands

Initial reproduction attempts on lower core counts timed out, blocking a full reproduction of 10000 OD pairs (some scenarios required a high-memory allocation of 48 GBs to complete within a reasonable timeframe of 6 hours).

3.3 3. Relative Path Logic and Execution Context

The authors' script used path logic (e.g., `os.path.join(os.getcwd(), '1_1_1')`) that assumed execution from within the source repository. While adhering to the authors' intended working directory would have resolved this, the associated script was redesigned to fit into the installed pipeline to ensure robust path resolution.

3.4 Recommendations for Future Reproducibility

1. Avoid Hardcoded Resource Counts: The associated generalizability to a machine-agnostic variable to limit allocated cores.
2. Robust Path Management: Use Python's `pathlib` for handling paths and path resolution.
3. Provide a "Dry" Dataset: Include a small subset of OD pairs (e.g., 100) to allow reviewers to verify the pipeline logic quickly without loading massive HPC resources.

4 Acknowledgements

This work used the Scientific Compute Cluster at GWU, the data center of Max Planck Society for the Advancement of Science (MPG) and University of Göttingen. In part funded by the Deutsche Forschungsgemeinschaft (DFG) German Research Foundation - 389577381.

5 Reproduced Figures

The reproduction pipeline successfully generated the following figure (only the first subset of 10000 OD pairs). The plots below were generated exactly as the automated verification pipeline.

5.1 Figure 1: Modal Split comparison of fastest path and average thresholds

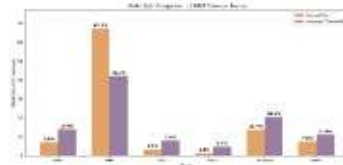


Figure 1: Modal Split Comparison (Figure 1)

Statistical Success: Reproduction Note: The figure was successfully reproduced. It confirms that the difference of 'average thresholds' (the include heading) significantly increases the Public Transport (PT) share and reduces the cycling share compared to a pure fastest path model.

5.2 Figure 2: Detailed modal split comparing the cycling restricted scenario (RT1250)

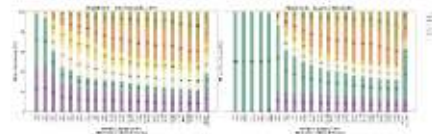


Figure 2: Cycling Restricted (RT1250) Comparison (Figure 2)

HPC, 279 core hours

Example reproducibility review reports from AGILE 2026



Reproducibility Review of: Maximizing Data Coverage: Fusing Street View and Oblique Imagery to Quantify Vertical Greenery Potential in Urban Areas



Item	Value
Title	Maximizing Data Coverage: Fusing Street View and Oblique Imagery to Quantify Vertical Greenery Potential in Urban Areas
Authors	Aruscha Kramm (0009-0002-7821-9388), André Ludwig, Bogdan Franczyk
Ref. paper	Kramm, A., Ludwig, A., and Franczyk, B.: Maximizing Data Coverage: Fusing Street View and Oblique Imagery to Quantify Vertical Greenery Potential in Urban Areas, AGILE GIScience Ser., 7, 8, https://doi.org/10.5194/agile-giss-7-8-2026 , 2026.
Codechecker's date of check	JeongWan Choi (0009-0002-9227-1151) 2026.01.13
Summary	Periodically reproducible
Repository	https://github.com/reproducible-agile/QuantifyVGS
Ref. certifiers	https://doi.org/10.5194/agile-giss-7-8-2026

Table 1: Reproduction metadata

Outputs and results

Running the workflow produced several plots illustrating distributions of façade characteristics and the calculated potential index. These include:

- façade orientation distributions
- window-to-wall ratio values
- façade surface area values
- calculated vertical greenery potential index values

The following figures were produced from the four-wall example dataset and demonstrate the execution of the downstream workflow. The reproduced plots are consistent with those produced by the provided notebook in the repository (calculable potential index plots).

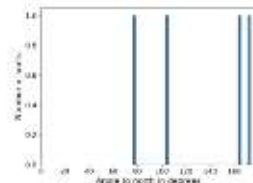


Figure 1: Orientation distribution (reproduced plot).

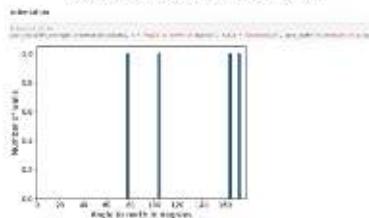


Figure 2: Distribution distribution (reproduced plot).

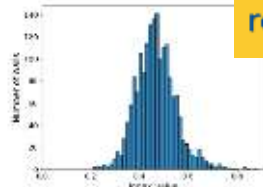


Figure 3: The calculated index for all walls in the test area (original plot in paper).

Figure 4 (renewed) is the output of the reproduced pipeline based on the provided sample dataset (four walls). As the corresponding figure in the paper (Figure 3) is based on the full dataset, a direct comparison is not possible; therefore, the reproduced results demonstrate correct execution of the workflow rather than replication of the published results.

Communication with the author

The author was helpful and responsive during the reproducibility assessment. After identifying an issue in the preprocessing script, the author promptly corrected the problem and updated the repository. Also, the author provided additional intermediate sample data and detection outputs, which enabled verification of the downstream computations shown.

Remaining limitations

Full end-to-end reproduction of the workflow from raw inputs was not possible because several components depended on non-public resources. In particular, the street-view and oblique image acquisition workflow, proprietary imagery datasets, and trained object-detection models used for façade element detection are not publicly available. The workflow also relies on API-based data acquisition, which requires restricted credentials. Scripts for working with uncorrelated credentials are not available in the repository, thereby limiting the reproducibility of the data acquisition step. Therefore, it was unable to reproduce the entire pipeline independently of the raw inputs.

Recommendations

Based on this, could be further enhanced by the following ways:

- Provide software specifications such as requirements on environment (e.g. OS).
- Include a minimal runnable sample dataset, covering all publicly accessible stages of the workflow.

licensed data > example data; fix in preprocessing script > partially reproducible

Reproducibility reviews in AGILE-GISS proceedings



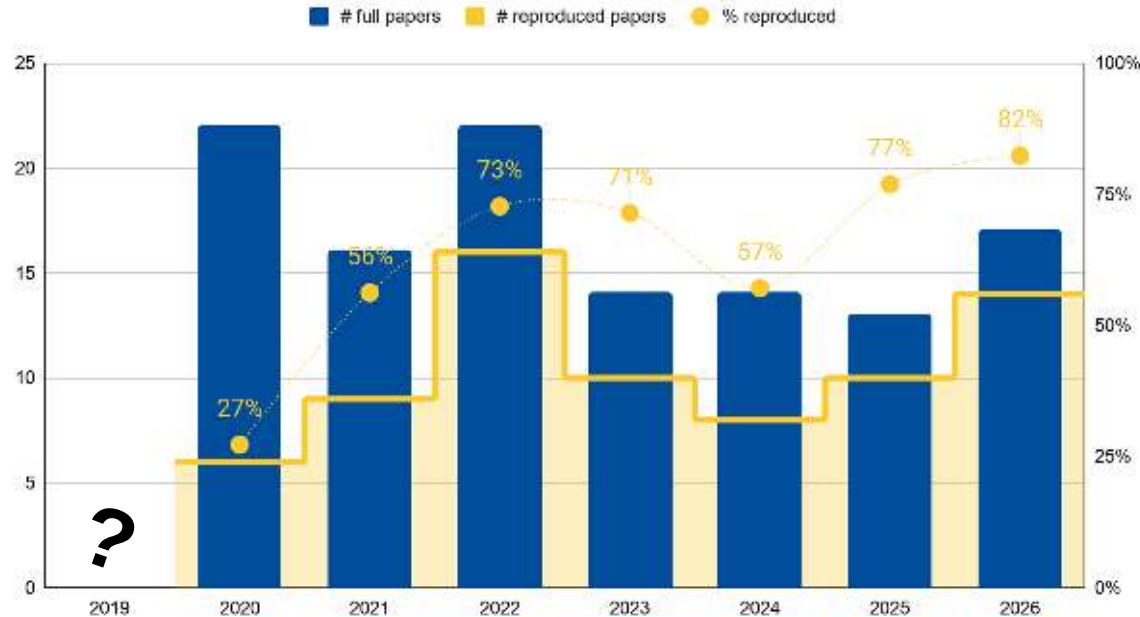
A collage of various pages from the AGILE-GISS proceedings. The top left page is the "Volume 7, 2026" cover. Below it is the "29th AGILE Conference on Geographic Information Science" title page. To the right is a "Table of Contents" page. In the center is an "Article" page titled "Analysis of the spatio-temporal pattern of COVID-19 incidence rate across Germany". To the right of the article is a "Full paper" page. At the bottom left is a "Geo-Agility of Urban Cognitive Regions" page. At the bottom center is a "Geo-Agility of Urban Cognitive Regions" page. At the bottom right is a "Geo-Agility of Urban Cognitive Regions" page. The pages are arranged in a way that shows the flow of the proceedings, from the cover to the table of contents, then to individual articles and full papers.

<https://agile-giss.copernicus.org/>

Reproducibility review results



Reproduced full papers @ AGILE 2019 to 2026



Example reasons for not reproduced:

- unresponsive authors (timing, availability)
- technical challenge (old code)

<https://osf.io/hmxda>
<https://doi.org/10.53962/nxah-m4>

How to put your community on a path towards more reproducibility in 5 easy hard steps

1. Build a team of enthusiasts (workshop, social events)
2. Assess the current state and raise awareness (workshop, paper)
3. Institutional support ([AGILE Council](#) + committee chairs)
4. Positive encouragement (no reproduction != bad science)
5. Keep at it!

Future



Do it again in 2027

Continue discourse on path towards Open Scholarship

Review/Publish computational notebooks?

Require CRediT?

Reviewer activity @ ORCID?

Open review, if reviewer is tenured?

Read our reports and more ideas [here](#).

Improve publishing workflow, depositions of records (open research information), and improve credit

Improve communication (e.g., no badge != bad)

Review paper 2017/18: How did we examine AGILE papers' potential reproducibility?

Collect AGILE
best papers

32 (20 full, 12
short)
papers from
2010 - 2017
(8% of 253 full
papers since
2007)

Assign 2
reviewers per
paper

Conceptu
al paper?

Y

Exclude from
further analysis

N

Assess
reproducibility

Reviewer
s agree?

N

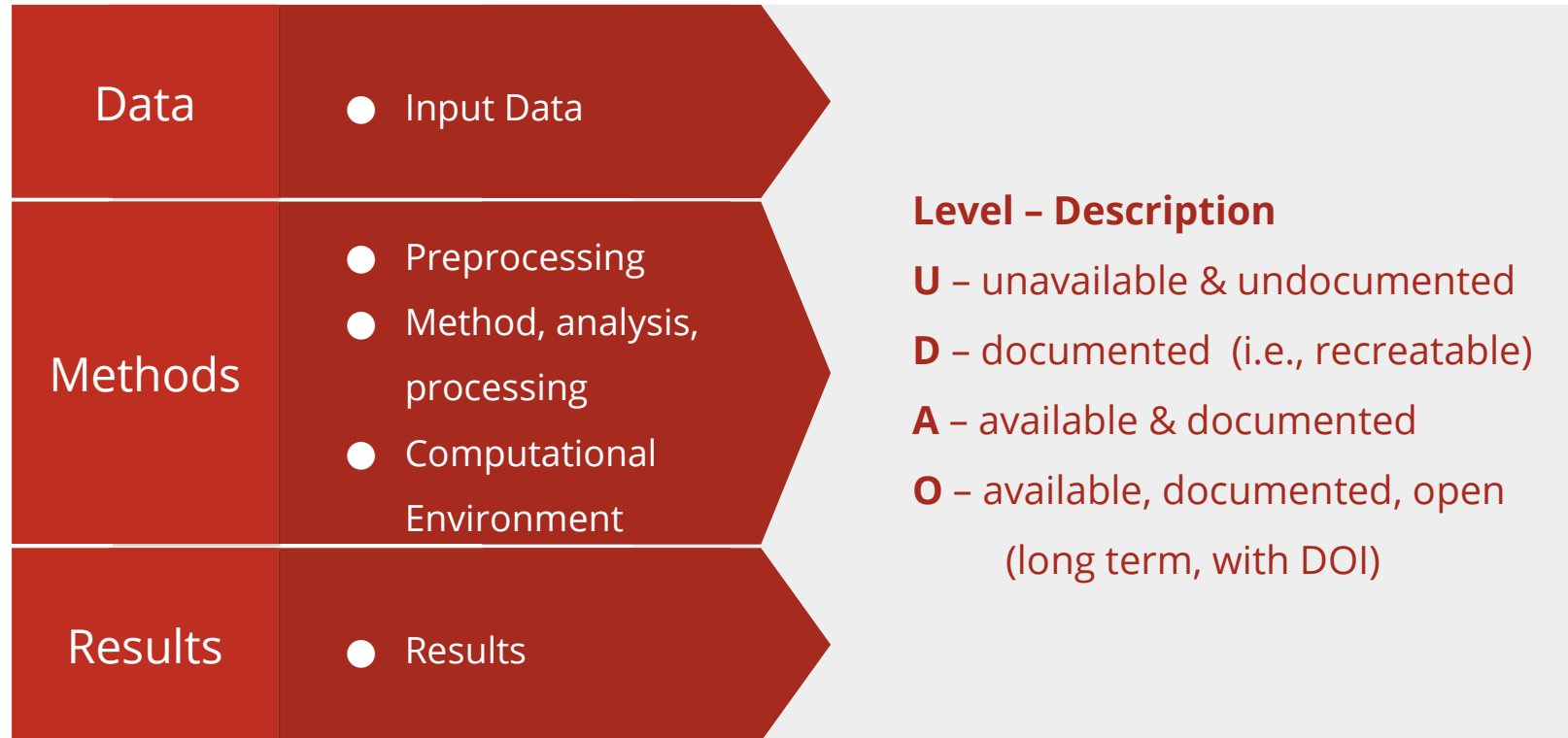
Discussion and
ultimately vote

Y

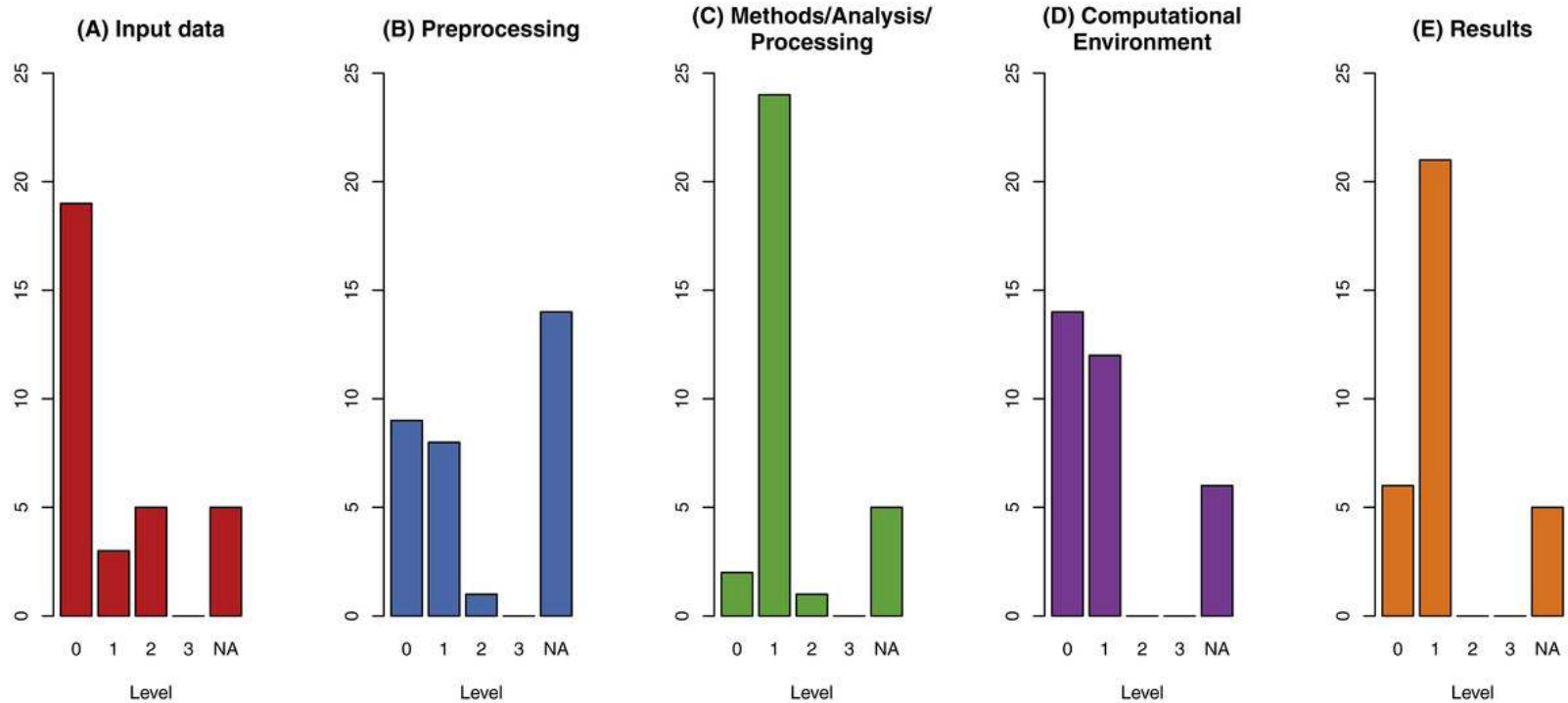
Analyze and
interpret
outcomes

Author survey

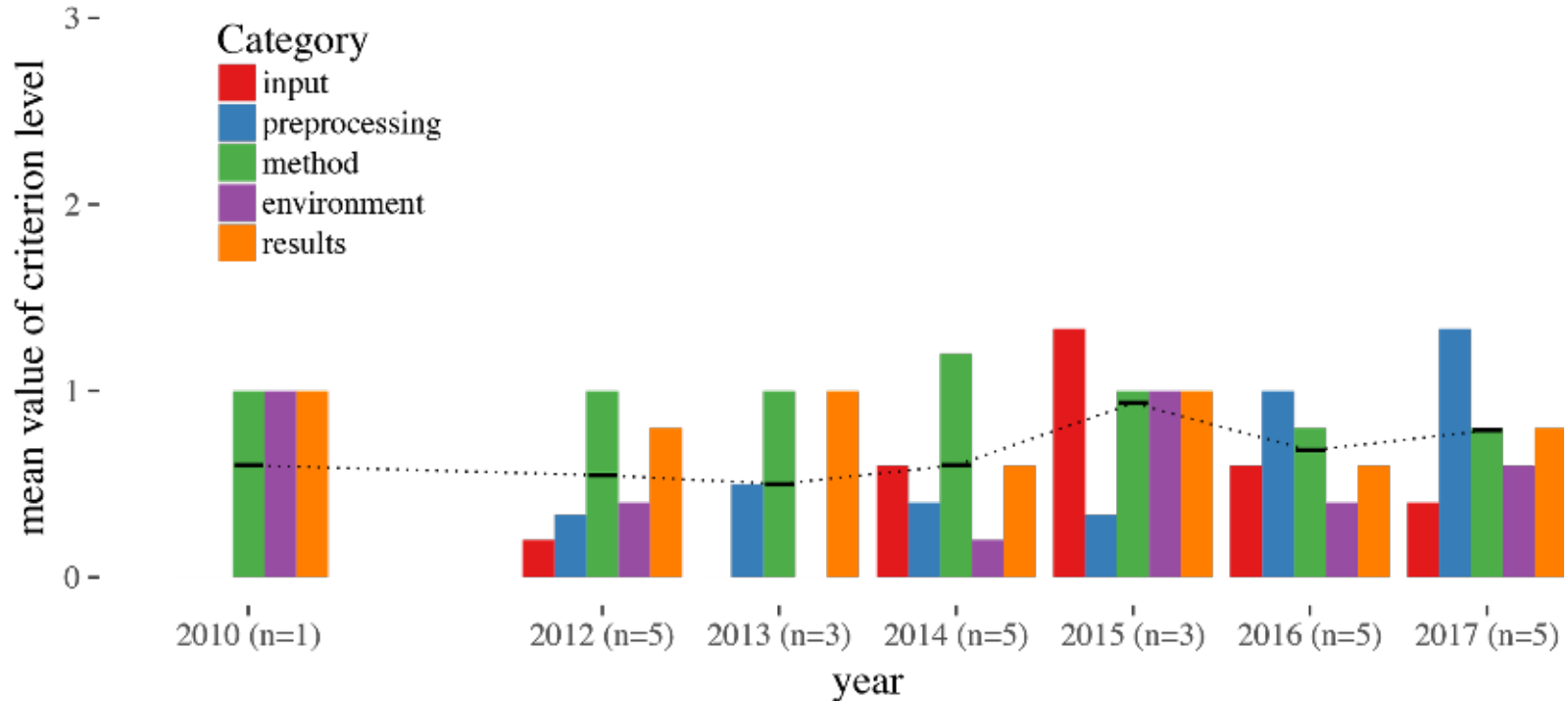
How to assess reproducibility?



How reproducible were AGILE papers?



Does it at least improve over time? (no)

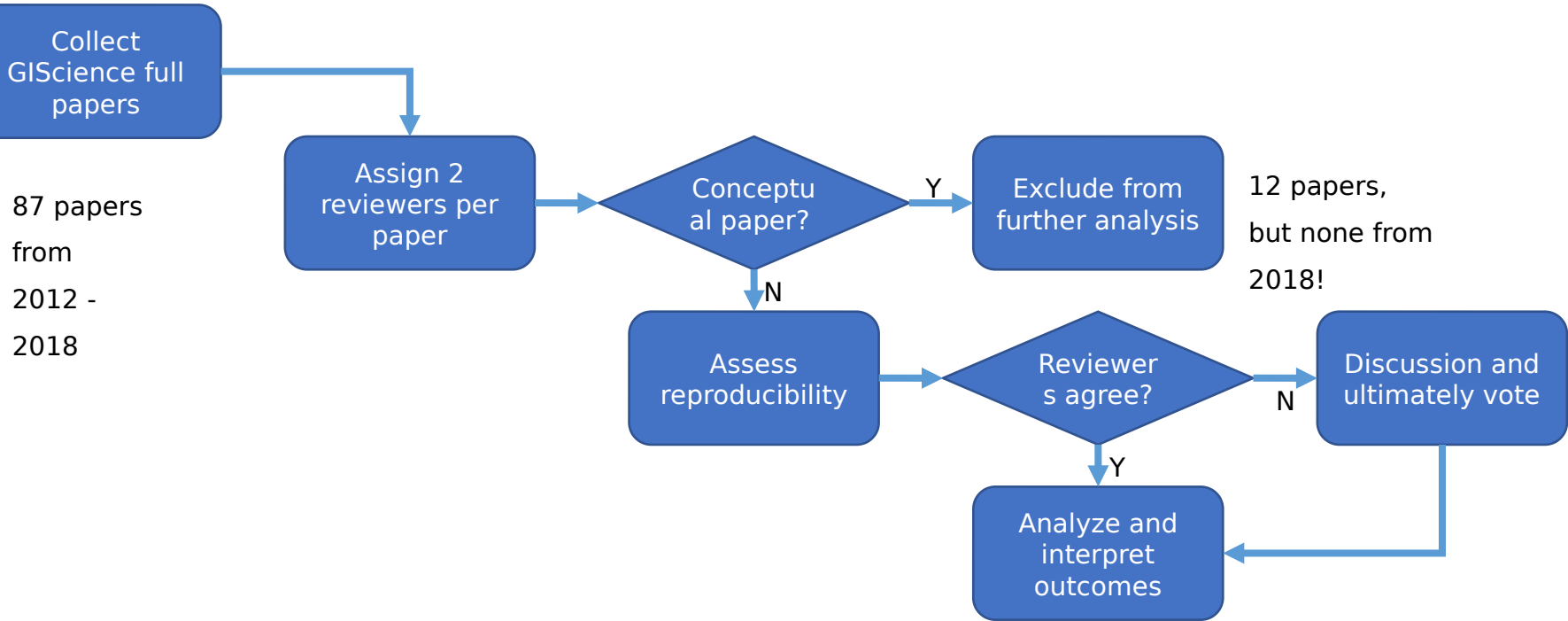


Next step: Replicate with another conference

What did we want to do?

1. Investigate the state of reproducibility at GIScience conference series
2. Replicate an earlier assessment for AGILE conference series:
 - Is the method generalizable?
 - How do AGILE and GIScience compare?
3. Discuss strategies for improving reproducibility

How did we go about it?



Was our approach replicable?

Short answer: yes

But:

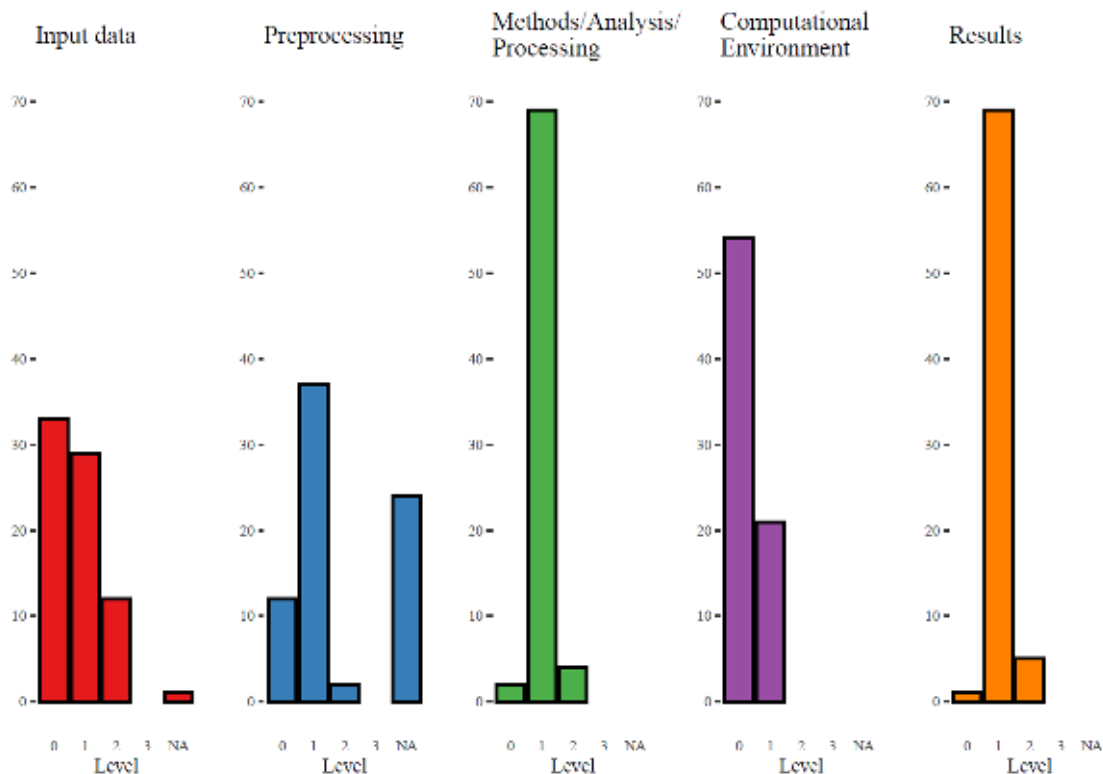
- labor-intensive, thus difficult to scale up
- *Preprocessing* not too helpful criterion (overlap with *Analysis*)
- *Computational environment* of limited use because relates mostly to processing time

Future replications should drop *preprocessing* and could drop *computational environment* criteria

Try it out!

<https://github.com/nuest/reproducible-research-at-giscience>

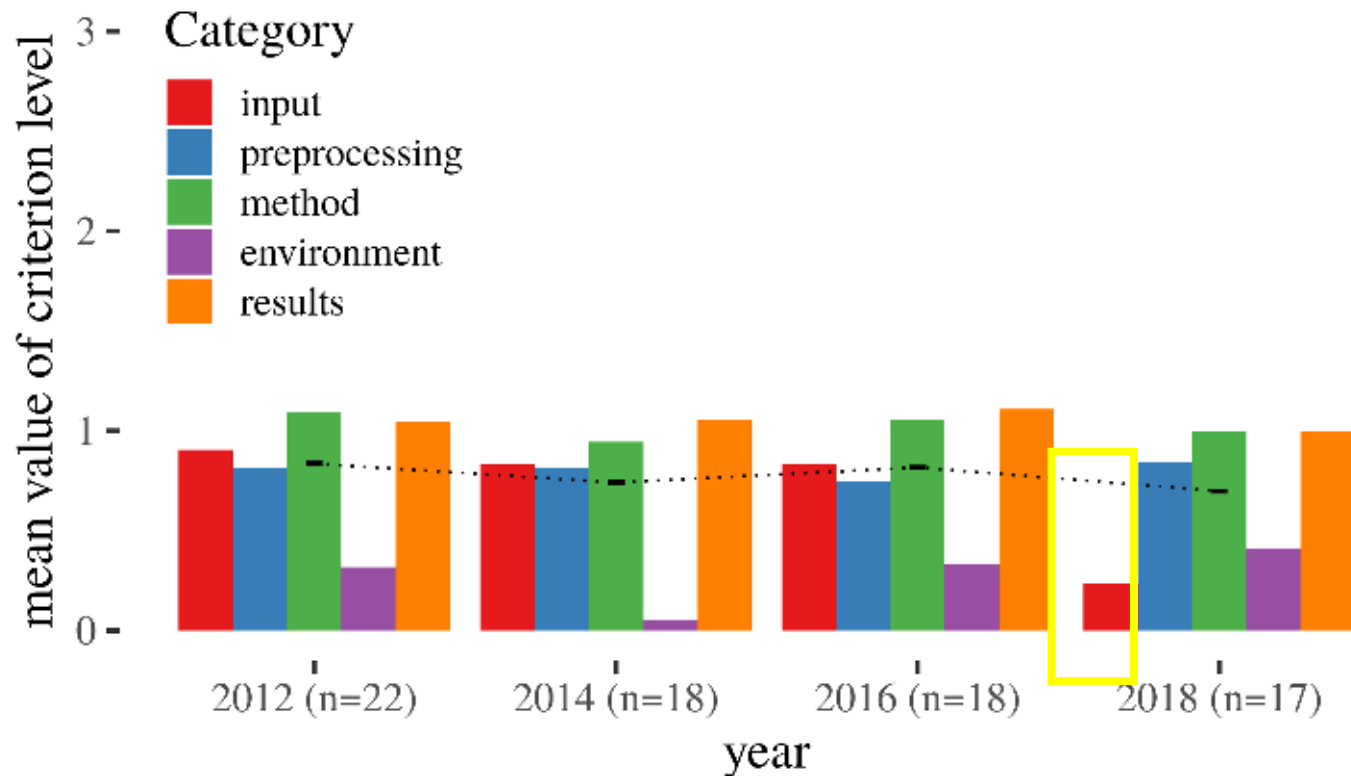
What's the outcome for GIScience?



Level –
Description
0 – unavailable
1 – documented
2 – available
3 – available and open

Figure 1 Barplots of reproducibility assessment results; levels range from 0 (leftmost bar) to 'not applicable' (rightmost bar).

Any change over time? (again, no)



But what does this mean for GIScience?

- Overall reproducibility not great but: most papers meet standards for publication (*'documented'* in all three main criteria)
- Main problem is *input data* (several score only *'unavailable'*)
 - Scores not a result of link rot (although that is a problem!): if there was reason to assume data was available at time of publication, paper received *'available'*
 - Worrisome, because of increased focus on data science and need for ML training data

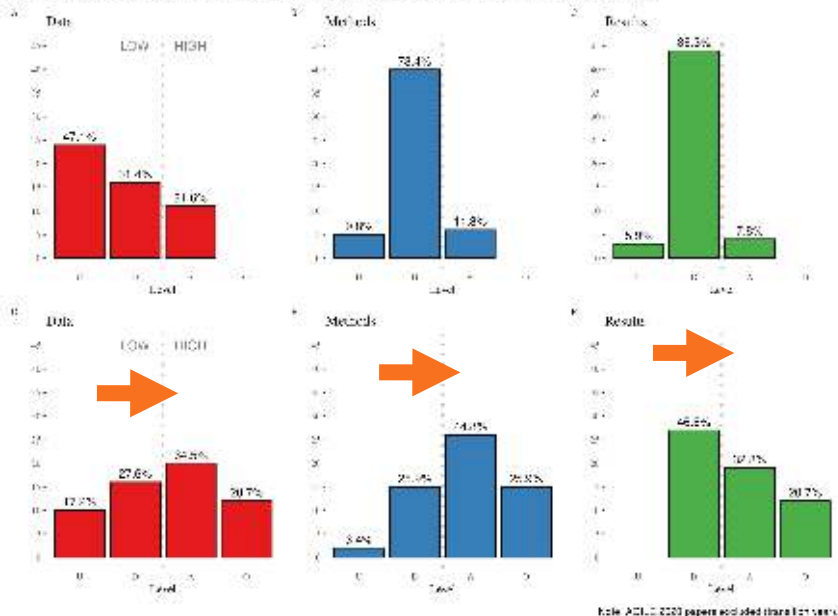


Results indicate that the reproducibility guidelines and the review process measurably improved the potential reproducibility of AGILE publications. [...] clear and enforced guidance is a key driver for change.

AGILE paper assessment according to the criteria 'Data', 'Methods', and 'Results'

Top: pre intervention (2017-2019) papers (N=51). Bottom: post intervention (2021-2024) papers (N=59).

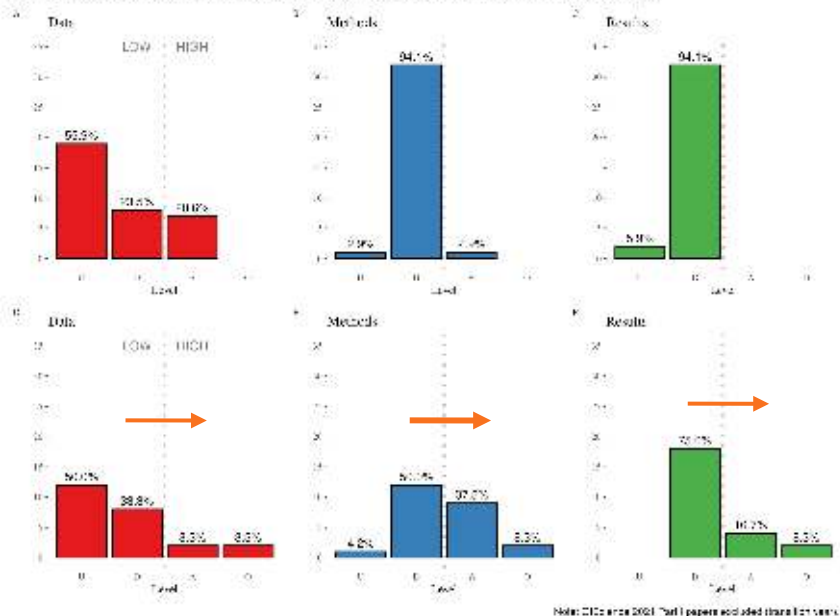
X-axis: axis: (U)ndocumented, (D)ocumented and (A)vailable Documented, Available, Available and (O)ther.



GIScience paper assessment according to the criteria 'Data', 'Methods', and 'Results'

Top: pre guidelines (2016-2019) papers (N=84). Bottom: post guidelines (2021-2024) papers (N=84).

X-axis: axis: (U)ndocumented, (D)ocumented and (A)vailable Documented, Available, Available and (O)ther.



Granell, C., Ostermann, F. O., Nüst, D., Kedron, P., Koukouraki, E., Matey-Sanz, M., Decoupes, R., Trilles, S., Graser, A., & Niers, T. (2026). Improving reproducibility of GIScience publications through novel reproducibility guidelines and revised review procedures. *Journal of Spatial Information Science*, (32), 67-93. <https://doi.org/10.5311/josis.2026.32.591>

It works...

... for **AGILE**:

1. [...] the odds of a paper scoring a higher reproducibility rank are significantly different in post-intervention AGILE proceedings [...], after accounting for author non-independence.
2. The OR are large and positive, [...], suggesting that post-intervention AGILE papers are significantly more likely to reach higher reproducibility levels.

... and for GIScience:

3. [...] for two criteria. The odds of a paper scoring a higher reproducibility rank are significantly different in post-intervention GIScience proceedings [...] for the Methods and Results criteria, but not for Data, after accounting for author non-independence.
4. [...] the OR are clearly positive for Methods and Results, so the change is towards higher ranks for the post-intervention group.

We observe overall **notably lower OR** for improved potential reproducibility **at GIScience** [...], while **AGILE shows significant and strong improvements for all criteria**. Combined with the fact that many post-intervention GIScience papers have at least one (co-)author who published also at AGILE, we are confident to state that, based on the statistical testing, the improvement of potential reproducibility at AGILE is larger and broader than the improvement at GIScience.



Criteria	p-value	OR	logOR
Data	<0.0001	19.271	2.959
Methods	<0.0001	20.485	3.020
Results	<0.0001	12.333	2.512

Table 3: Differences between pre- and post-intervention periods for AGILE conference. OR: odds ratios. Source: 04_results_hypotheses.ipynb [16].

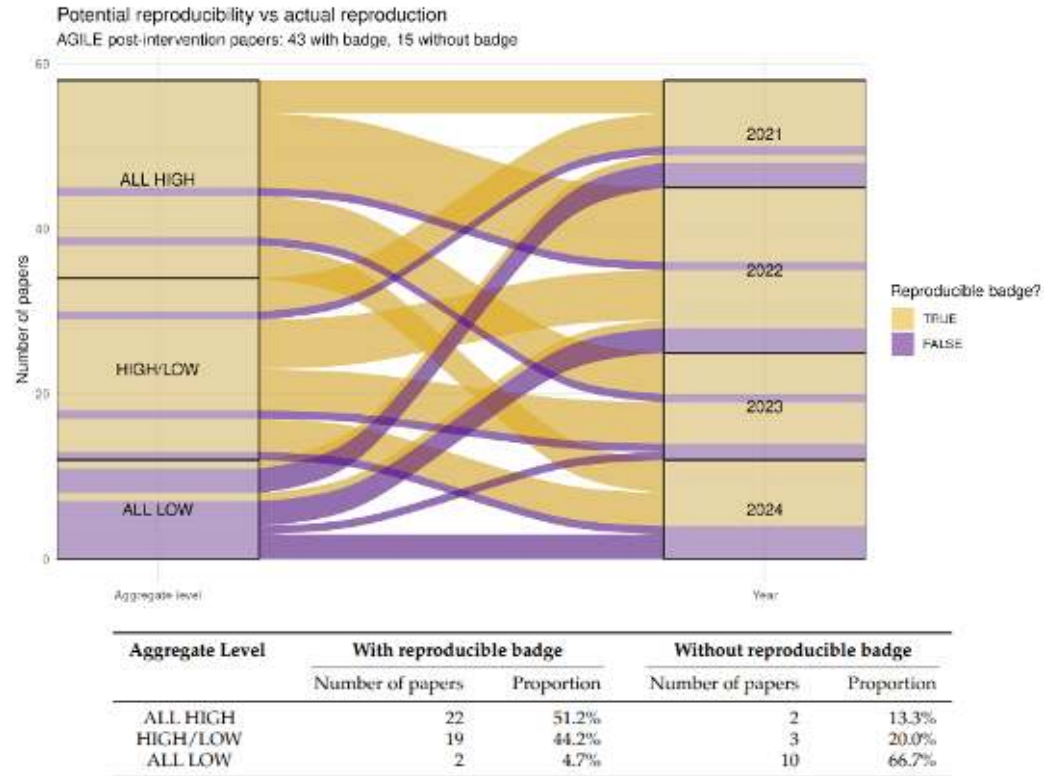
Criteria	p-value	OR	logOR
Data	0.0683	1.995	0.691
Methods	<0.0001	8.365	2.124
Results	<0.0001	5.329	1.673

Table 4: Differences between pre- and post-intervention periods for GIScience conference. OR: odds ratios. Source: 04_results_hypotheses.ipynb [16].

Criteria	AGILE	GIScience
Data	33.6	-3.9
Methods	58.9	42.9
Results	45.7	25.0

Table 5: Absolute change in percentage points between GIScience vs AGILE after the intervention for the aggregate level *High*. Source: 03_results_reprolevels.qmd [16].

It works...



Granell, C., Ostermann, F. O., Nüst, D., Kedron, P., Koukouraki, E., Matey-Sanz, M., Decoupes, R., Trilles, S., Graser, A., & Niers, T. (2026). Improving reproducibility of GIScience publications through novel reproducibility guidelines and revised review procedures. *Journal of Spatial Information Science*, (32), 67–93. <https://doi.org/10.5311/josis.2026.32.591>

Figure 2: Potential reproducibility levels as indicators of reproduction in AGILE post-intervention papers. Figure shows connection between aggregated levels and reproducible badges earned. Levels *Undocumented* and *Documented* on all criteria are considered *All Low*, levels *Available* and *Open* on all criteria are considered *All High*. Table below shows actual numbers and proportions for context. Source: 06_discussion.qmd [16].

From AGILE to... IPIN?

Ideas that are transferable to any community-led research conference: human-centered process, communicative focus, supportive framework of tools and infrastructure.





Human-centered process for reproducibility handles variety

Author provides material and makes fixes

Collaboration makes things work

Codechecker stops when confident enough all parts are provided and then write review

Aligns with peer review practices

But: challenge of codechecker matching and complex procedures?

Communicative focus.

Avoids rules & automation playing catch with innovation & technology
Avoids unification or limitation of researcher freedom

Continuous development of shared practice over time
& definition of “reproducibility” and “how reproducible is enough”

Targets attitudes towards sharing, transparency, and openness,
focuses on positive learning experience for all involved roles

Tools & infrastructure... with care.

Faster codechecking with...

Notebooks

Binder (BinderHub)

Open methods

Missing?!

- free Binder-ready notebooks close to the data
- metadata connectivity > credit & partial re-building
- (small) reference datasets & demo pipelines

A man with a beard, wearing a dark blue t-shirt, is looking at a row of vending machines. The machines are filled with various snacks, drinks, and instant noodle cups. The scene is dimly lit, with the machines providing the primary light source. The text "Where to start, what to pick?" is overlaid in white on the left side of the image.

Where to start,
what to pick?.



<https://media.giphy.com/media/mbhseRYedIG5W/giphy.gif>

What are the steps that you could take?

Initiate discourse with a core team in a
(small, sub) community (e.g., small event, journal)
Document state of reproducibility &
find leadership support
Establish guidelines (do-ocracy) &
apply to current/historic work (post publication?)
Continue development and positive discourse
Share experiences and publish (ORI)

mind set



Challenge: For next group meeting

- Come up with three reasons why Open Science is a *bad* idea!
- Match codecheckers to upcoming papers (pre-publication)
- Find one paper from the extended materials below and present it

Today is the first day of the rest of your reproducible career!

What about Generative AI?

Massive challenge for computational reproducibility itself
(explainable AI, ...)

Opportunity: good workflows “for free”
(tests, documentation, environments, CI)

Assistant in checks (small fixes on the way, writing, metadata)

Where should humans take over? They must take responsibility.

Lorena Barba Reproducibility in the Age of Agentic AI: Context Engineering at the Timescale of a Codebase
<https://arxiv.org/abs/2609.11728v1>



Code checking at IPIN 2026+

Started the conversation

First steps taken with guidelines and **compliance testing** after acceptance, boldly take the next step next year and start **executing workflows**

Meta-reviewers & TPC = excellent set-up > could consider reproducibility review

Innovate (open hardware? reproducible experiments?)

Educate (e.g., ECRs, workshops, ..)

Integrate (e.g., publisher collaboration, metadata/ORI)

Second attempt not ready yet - excursion: Codechecking with Claude

Prompt

we act as a codechecker for the PDF in the project folder, see <https://codecheck.org.uk/guide/community-workflow-codechecker> for our tasks;

note the check is unsolicited - the author did not prepare any workflow; we use the Quarto-based template from <https://codecheck.org.uk/codecheck/>;

the code is already downloaded from the Zenodo repository at <https://doi.org/10.5281/zenodo.21839295> to a local folder;

extract information from the PDF to fill in the codecheck.yml file and analyse the workflow in "database & code" - identify the main control file and dependencies first;

Response

I've **filled in codecheck.yml** and analysed the workflow in database & code/. Nothing has been run yet. The most important finding is that **the code as published can't produce any results, because of three bugs that stop it before any output is written. The other main gap is that the authors include no script that turns the raw results into Tables I and II or Figure 1.**

...

Codechecking with Claude (output cont.)

codecheck.yml

- **Paper:** I took the title and both authors from the PDF and the ORCIDs from the Zenodo record: NN, NN.
- Code and data: the Zenodo DOI goes in repository, and a source entry describes the zip. The local zip matches the Zenodo md5 checksum.
- **Manifest:** since the check is unsolicited, it lists the files we plan to create: NN.
- **Codechecker:** I entered you as ...
- Still **FIXME:** the paper's DOI (none yet for IPIN 2026), the certificate ID, the report DOI, the summary and your own repository URL. The YAML syntax is valid. The R validator's only error is the expected one about the placeholder certificate ID (2026-XYZ).

How the workflow fits together

The code is MATLAB/GNU Octave. The .mat files were made in MATLAB on Windows, but the experiments are run with octave --eval.

1. Data preparation: ... The step that makes the ... files is not included.
2. Generating the run scripts: ...
3. Main control file: ...
4. Per-run code: ... loads the data with ... Each run writes...csv.
5. Missing step: there is no code that turns those ... tables or Figure.

Dependencies: GNU Octave is the only one; there are no toolboxes or packages.

Neither Octave nor MATLAB is installed here, but Docker is, so the gnuoctave/octave image is the obvious environment.

Codechecking with Claude (output cont.)

Problems found while reading the code

Stop the code from running:

- Method names don't match the files. ...
- Double file extension. ...
- Crash at the end of every run. ... has 3 columns, but the summary printout reads columns 4–6. ...

Paths and setup:

- Wrong location for the scripts. ...
- Missing folder. There is no NN/ directory, and the redirect ... fails without it.

Mismatches between the code, README and paper:

- Wrong argument order in the README. ...
- Datasets that don't exist. ...
- Typo. ...
- Different k values for the averaged set. ...

...

Matches with the paper: ...

Suggested next steps

1. Set up Octave in Docker.
2. Make a working copy with the minimal fixes for the three blocking problems and record them in the certificate. The guide says the codechecker "records but does not fix", so this needs to be documented. Contacting the authors is optional.
3. Run a single configuration to time it, then decide ... can't be identified ... unless the authors share their selected hyperparameters; the paper mentions publishing them, but they are not in the Zenodo record.
4. Write our own script to recompute Tables ... Figures.

Shall I go ahead with steps 1–3?

Codechecking with an AI assistant

Quickly identified problems - sound reasonable

Typical (no blame!) **situation:**

- README not sufficient

- Workflow likely “cleaned” for publication but not “actual workspace”

Next:

- Draft email to authors

- Create local rules file (CLAUDE.md) to get details about the process right

A large crowd of people, mostly men, are seated in a stadium with bright yellow seats. They are wearing various casual clothing like t-shirts, tank tops, and hats. The scene is brightly lit, suggesting an outdoor daytime event. Overlaid on the center of the image is the text "Open Science Open Research" in a large, white, sans-serif font. The words "Open Science" are on the top line, and "Open Research" is on the bottom line, with the two lines slightly offset from each other.

Open Science Open Research

Learn crafts by doing.

Be kind. Help. Share.

Be open.

Thanks to Frank Ostermann and Markus Konkol - some slides are based on their presentations, e.g., for “M-SE Data Mastery - Open and Reproducible Research”.

Thanks!

**What are your thoughts
and questions?**

more more

more more

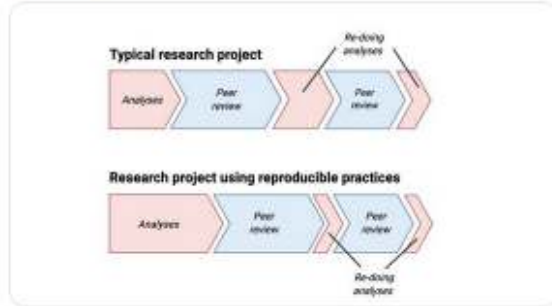
more more



Dan Quintana
@dsquintana

In my experience, you don't lose time doing reproducible science—you just «relocate» how you're spending it

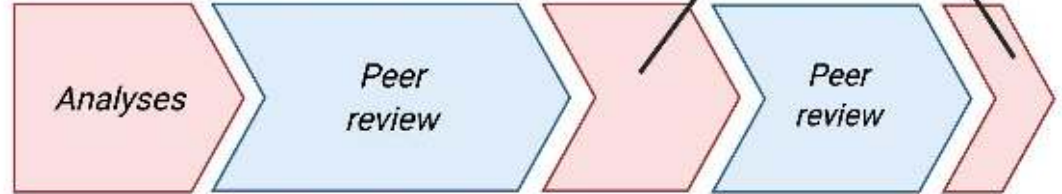
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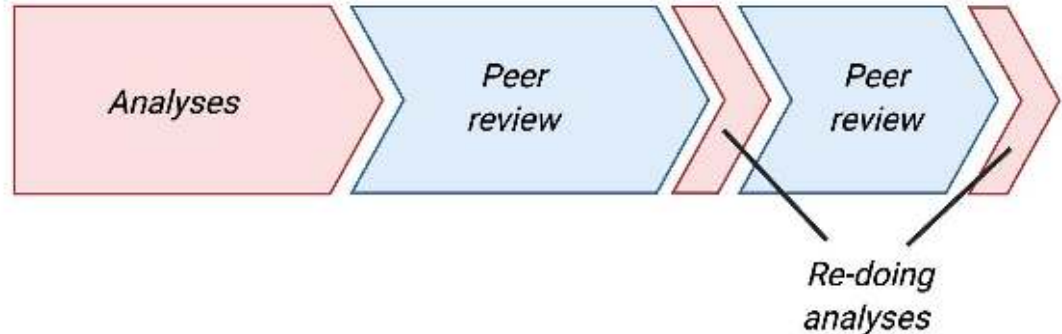
4:13 nachm. · 26. Nov. 2020 · TweetDeck

107 Retweets · 20 Zitierte Tweets · 536 „Gefällt mir“-Angaben

Typical research project



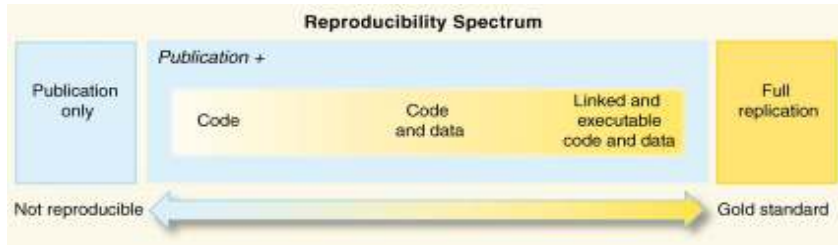
Research project using reproducible practices



Quintana, D. S. (2020, November 28). Five things about open and reproducible science that every early career researcher should know.

<https://doi.org/10.17605/OSF.IO/DZTVQ>

Reproducible Research & Open Science



<https://doi.org/10.1126/science.1213847>



		Data	
		Same	Different
Analysis	Same	Reproducible	Replicable
	Different	Robust	Generalisable

<https://the-turing-way.netlify.app/reproducible-research/overview/overview-definitions.html>



"Science should be 'show me', not 'trust me'; it should be 'help me if you can', not 'catch me if you can'."

Rather than reproducibility, should we be looking at preproducibility? @Nature wellc.me/2IMNuiq

151 15:55 - 28. Mai 2018



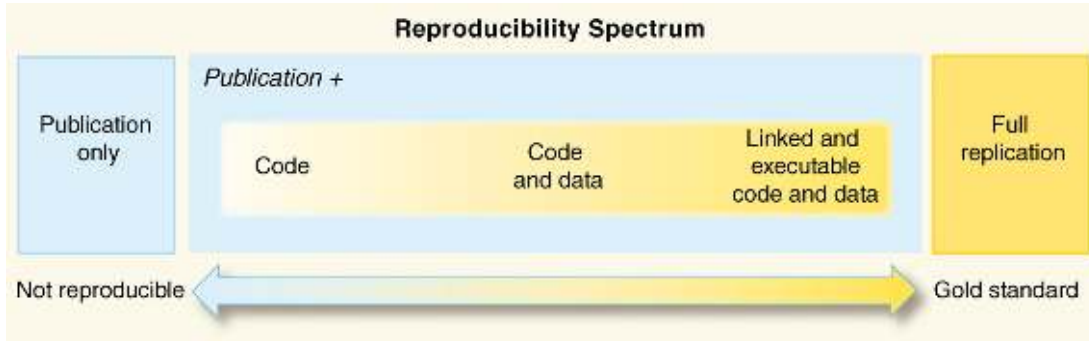
"Science should be 'show me', not 'trust me'."

Preproducibility

Before reproducibility must come preproducibility
Instead of arguing about whether results hold up, let's push to prov...
nature.com

<https://doi.org/10.1038/d41586-018-05256-0>

Reproducibility Spectrum & Preproducibility



<https://doi.org/10.1126/science.1213847>

Acknowledge spectrum and
(needed) mindsets

Individual solutions

Communication & humans



"Science should be 'show me', not 'trust me'; it should be 'help me if you can', not 'catch me if you can'."

Rather than reproducibility, should we be looking at preproducibility? @Nature wellc.me/2IMNuiq

151 15:55 - 28. Mai 2018



<https://www.nature.com/articles/d41586-018-05256-0>

What are reproducibility and replicability?

“**Reproducibility**, closely related to **replicability** and **repeatability**, is a major principle underpinning the scientific method. For the findings of a study to be reproducible means that results obtained by an experiment or an observational study or in a statistical analysis of a data set should be achieved again with a high degree of reliability when the study is replicated. There are different kinds of replication^[1] but typically replication studies involve different researchers using the same methodology. Only after one or several such successful replications should a result be recognized as scientific knowledge.”

([Wikipedia](#))

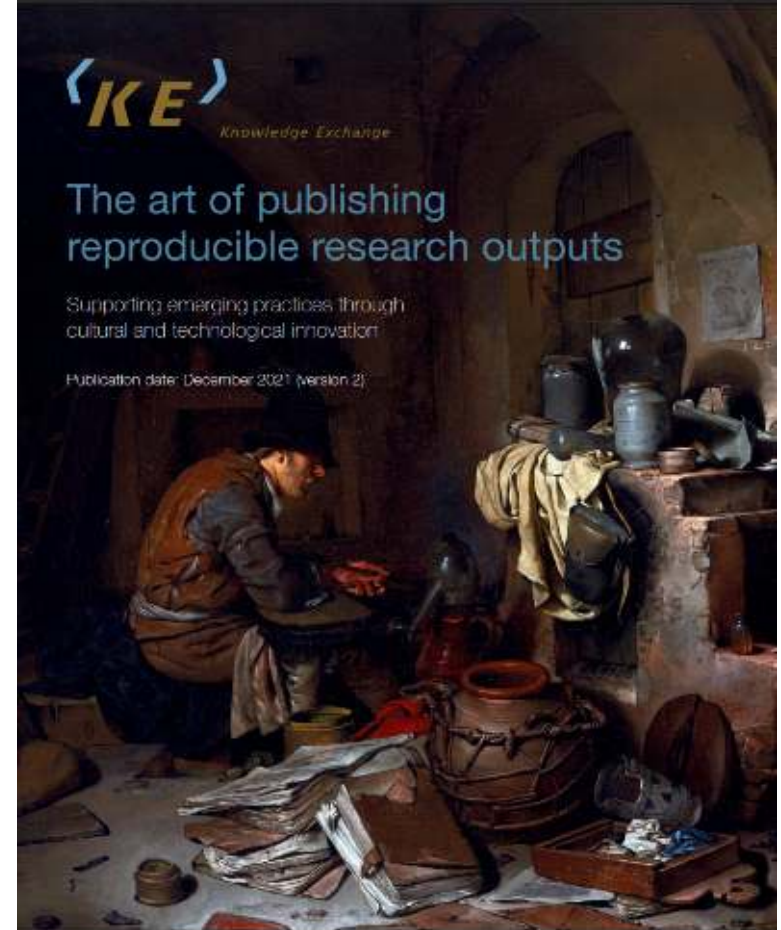
Knowledge Exchange

T&F group: Reproducible Research Outputs

<https://knowledge-exchange.info/event/publishing-reproducible-research-output>

Chiarelli, A., Loffreda, L., & Johnson, R. (2021). The Art of Publishing Reproducible Research Outputs: Supporting emerging practices through cultural and technological innovation. Zenodo.

<https://doi.org/10.5281/zenodo.5521077>





Professionalisation

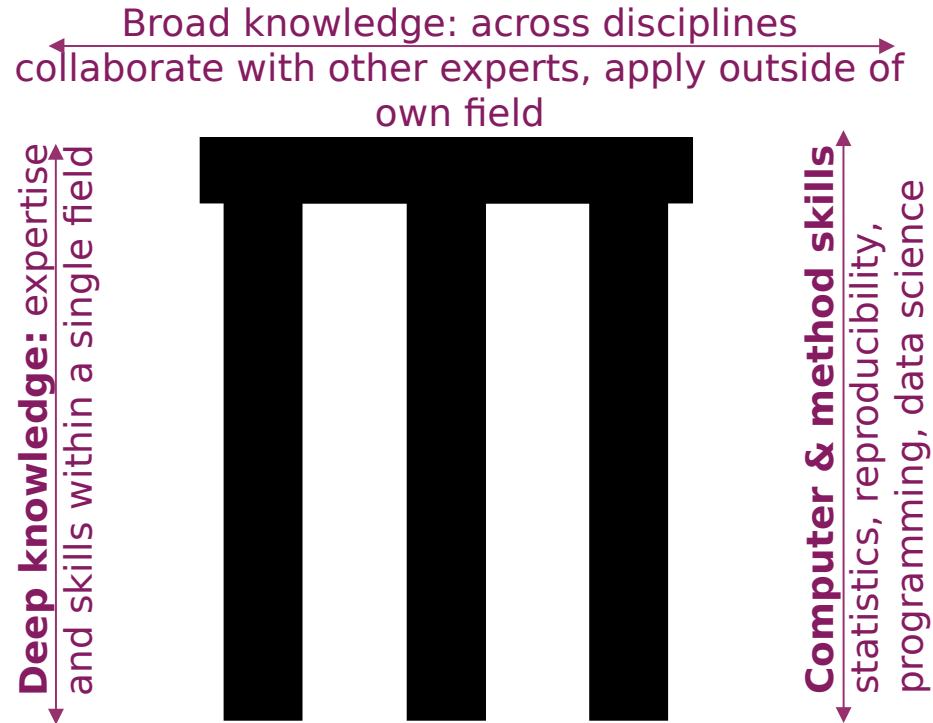


≠



BETTER infrastructu
SOFTWARE by ~~R~~SEs
BETTER
RESEARCH

Traditional scientists and modern eScientists



https://en.wikipedia.org/wiki/T-shaped_skills

<https://doi.org/10.1007/s10816-015-9272-9>

<https://jakevdp.github.io/blog/2014/08/22/hacking-academia/>

<https://www.sciencemag.org/careers/2013/05/when-all-science-becomes-data-science>

<https://escience.washington.edu/community-level-data-science-and-its-spheres-of-influence-beyond-novelty-squared/>

Literate Programming

Donald E. Knuth

Computer Science Department, Stanford University, Stanford, CA 94305, USA

The author and his associates have been experimenting for the past several years with a programming language and documentation system called WEB. This paper presents WEB by example, and discusses why the new system appears to be an improvement over previous ones.

A. INTRODUCTION

The past ten years have witnessed substantial improvements in programming methodology. This advance, carried out under the banner of "structured programming," has led to programs that are more reliable and easier to comprehend; yet the results are not entirely satisfactory. My purpose in the present paper is to propose another motto that may be appropriate for the next decade, as we attempt to make further progress in the state of the art. I believe that the time is ripe for significantly better documentation of programs, and that we can best achieve this by considering programs to be *works of literature*. Hence, my title: "Literate Programming."

Let us change our traditional attitude to the construction of programs: Instead of imagining that our main task is to instruct a *computer* what to do, let us concentrate rather on explaining to *human beings* what we want a computer to do.

The practitioner of literate programming can be regarded as an essayist, whose main concern is with exposition and excellence of style. Such an author, with thesaurus in hand, chooses the names of variables carefully and explains what each variable means. He or she strives for a program that is comprehensible because its concepts have been introduced in an order that is best for human understanding, using a mixture of formal and informal methods that nicely reinforce each other.

I dare to suggest that such advances in documentation are possible because of the experiences I've had over the past several years while working intensively

assistants; and why? Because it seems to me that at last I'm able to write programs as they should be written. My programs are not only explained better than ever before; they also are better programs, because the new methodology encourages me to do a better job. For these reasons I am compelled to write this paper, in hopes that my experiences will prove to be relevant to others.

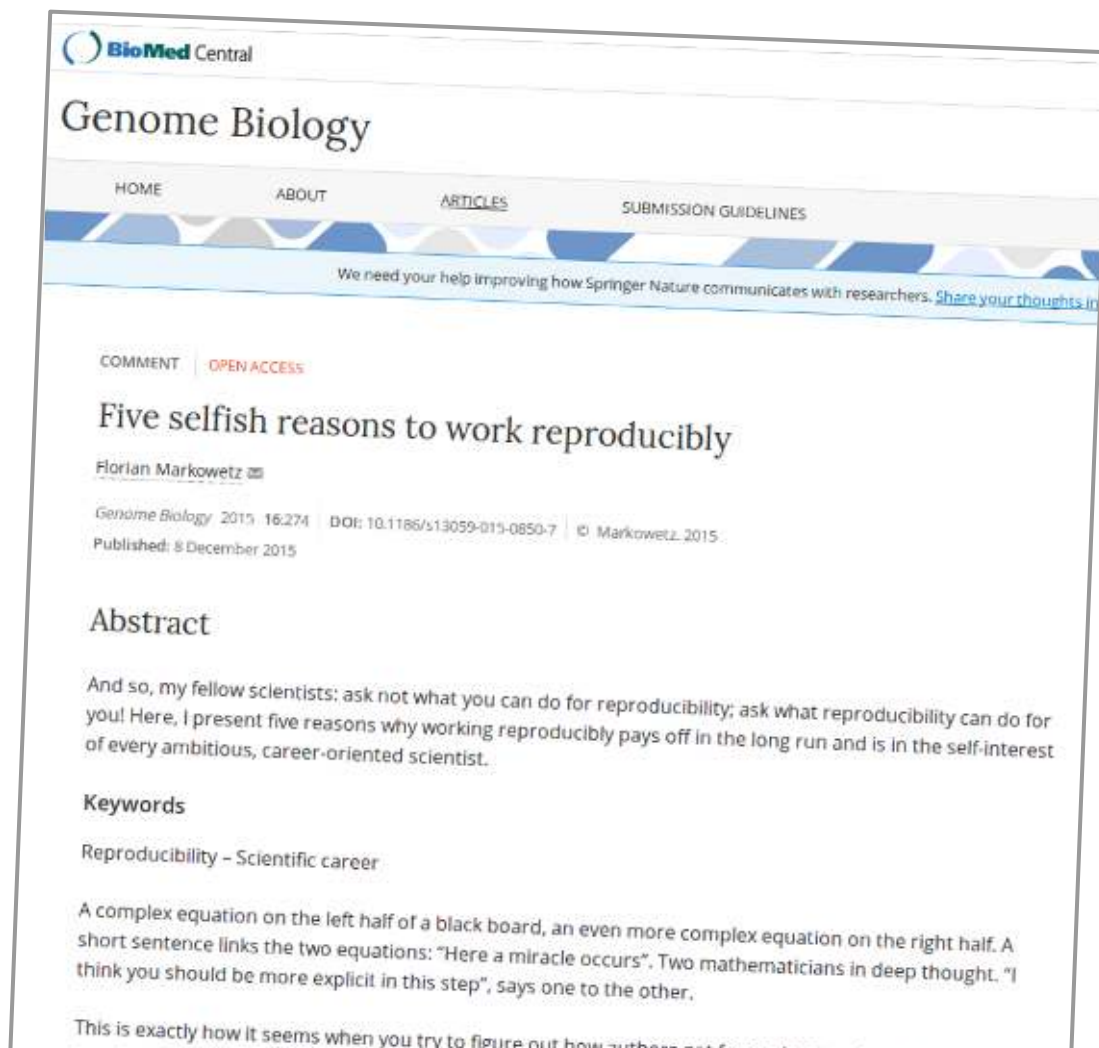
I must confess that there may also be a bit of malice in my choice of a title. During the 1970s I was coerced like everybody else into adopting the ideas of structured programming, because I couldn't bear to be found guilty of writing *unstructured* programs. Now I have a chance to get even. By coining the phrase "literate programming," I am imposing a moral commitment on everyone who hears the term; surely nobody wants to admit writing an *illiterate* program.

B. THE WEB SYSTEM

I hope, however, to demonstrate in this paper that the title is not merely wordplay. The ideas of literate programming have been embodied in a language and a suite of computer programs that have been developed at Stanford University during the past few years as part of my research on algorithms and on digital typography. This language and its associated programs have come to be known as the WEB system. My goal in what follows is to describe the philosophy that underlies WEB, to present examples of programs in the WEB language, and to discuss what may be the future implications of this work.

I chose the name WEB partly because it was one of the better words of English that hadn't already





Five selfish reasons to work reproducibly

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Five selfish reasons to work reproducibly

[Florian Markowetz](#) 

[Genome Biology](#) 16, Article number: 274 (2015) | [Cite this article](#)

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<https://doi.org/10.1186/s13059-015-0850-7>

1. reproducibility helps to avoid disaster
2. reproducibility makes it easier to write papers
3. reproducibility helps reviewers see it your way
4. reproducibility enables continuity of your work
5. reproducibility helps to build your reputation

4 things on Open Reproducible Research

Have a **README**: all else is details.

Have a **colleague** run your workflow before submission.

Reproduce papers (return the favour) to hone your **craft**.

Publish code and data, and **cite** it.

Open code for open science?

Steve M. Easterbrook

Nature Geoscience 7, 779–781 (2014) | doi:10.1038/ngeo2283
Published online 30 October 2014

Published online 30 October 2014



PDF



Citation

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Artículo matricial

Open source software is often seen as a path to reproducibility in computational science. In practice there are many obstacles, even when the code is freely available, but open source policies should at least lead to better quality code.

Subject terms: Climate and Earth system modelling

Poor code quality is endemic, and not just in scientific computation. It is always tempting to build something 'quick and dirty', under the assumption that it can be cleaned up later. This is especially true at the cutting edge of a field — why invest time writing beautifully engineered code from the outset, if you're not sure that what you're trying to do is even possible?

In software engineering, this is known as technical debt: by deferring issues such as code readability and maintainability, a debt is created that someone in the future might have to pay, in the extra effort needed to re-run or modify the code¹. The point of the metaphor is not that debt is bad *per se*. After all, we frequently incur debt to obtain something of immediate value, for example, using a mortgage to buy a house. The point is that such debts have to be managed carefully, to prevent them spiraling out of control.

Open source policies in scholarly journals can help here. If journals ask for open code, they create a strong incentive for authors to clean up the code each time a paper is produced, rather than deferring such tasks indefinitely. As a second order effect, such policies also

Editors' pick



Image credit: Johan Swanepoel / Alamy Stock Photo

Core Science 11

The Science Journal

Science events

natureevents directory

The 6th Hydrology, Ocean and Atmosphere
Conference (HOAC 2017)

16 June 2017 – 18 June 2017
Hangzhou, China

Visualizing Biological Data (VIZBI 2017)

14 June 2017 — 16 June 2017
The University of Sydney, Sydney, Australia

The 7th Int'l Conference on Geology and Geophysics (ICGG 2017)

20 July 2017 — 22 July 2017
Gullin, Gullin, Gullin, Ching

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Most read

A lower limit to atmospheric CO₂ concentrations over the next 100 years

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EDITORIAL

Ten Simple Rules for Reproducible Computational Research

Geir Kjetil Sandve , Anton Nekutenko, James Taylor, Elvind Hovig

Published: October 24, 2013 • <https://doi.org/10.1371/journal.pcbi.1003285>

Article	Authors	Metrics	Comments	Related Content
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Rule 1: For Every Result, Keep Track of How It Was Produced

Rule 2: Avoid Manual Data Manipulation Steps

Rule 3: Archive the Exact Versions of All External Programs Used

Rule 4: Version Control All Custom Scripts

Rule 5: Record All Intermediate Results, When Possible in Standardized Formats

Rule 6: For Analyses That Include Randomness, Note Underlying Random Seeds

Rule 7: Always Store

Citation: Sandve GK, Nekutenko A, Taylor J, Hovig E (2013) Ten Simple Rules for Reproducible Computational Research. PLoS Comput Biol 9(10): e1003285. <https://doi.org/10.1371/journal.pcbi.1003285>

Editor: Philip E. Bourne, University of California San Diego, United States of America

Published: October 24, 2013

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Funding: The authors' laboratories are supported by US National Institutes of Health grants HG005133, HG004909, and HG006520 and US National Science Foundation grant DBI 0850103. Additional funding is provided, in part, by the Huck Institutes for the Life Sciences at Penn State, the Institute for Cyberscience at Penn State, and a grant with the Pennsylvania Department of Health using Tobacco Settlement Funds. The funders had no role in the preparation of the manuscript.

Competing Interests: The authors have declared that no competing interests exist.

Top Tips to Make Your Research Irreproducible

Neil P. Chue Hong¹, Tom Crick², Ian P. Gent³, Lars Kotthoff⁴ and Kenji Takeda⁵

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²Department of Computing & Information Systems, Cardiff Metropolitan University, UK

³School of Computer Science, University of St Andrews, UK

⁴Insight Centre for Data Analytics, University College Cork, Ireland

⁵Microsoft Research, Cambridge, UK

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⁴lars.kotthoff@insight-centre.org <http://4c.ucc.ie/~larsko/>

⁵kenji.takeda@microsoft.com
<http://research.microsoft.com/en-us/people/kenjitak/>

1 April 2015

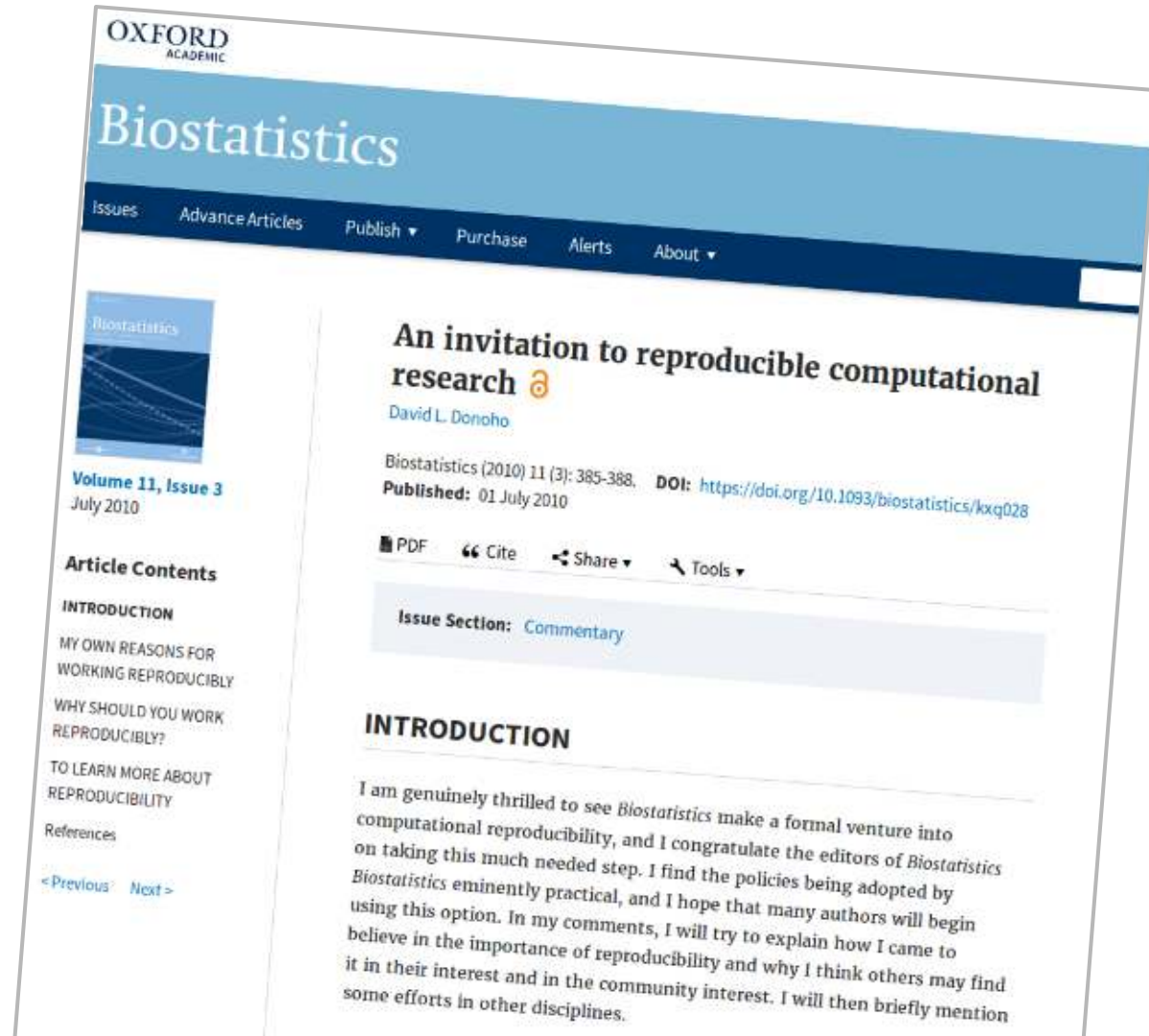
We have noticed (and contributed to) a number of manifestos, guides and top tips on how to make research reproducible [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]; however, we have seen very little published on how to make research *irreproducible*.

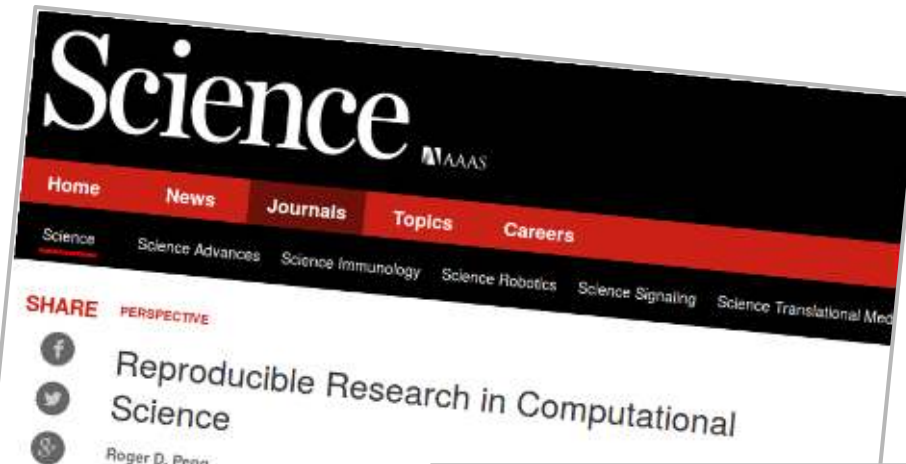
Irreproducibility is the default setting for all of science, and irreproducible research is particularly common across the computational sciences. The study of making your work irreproducible without reviewers complaining is a much neglected area; we feel therefore that by encapsulating some of our top tips¹ on irreproducibility, we will be filling a much-needed gap in the domain literature. By following our starter tips, you can ensure that if your work is wrong, nobody will be able to check it; if it is correct, you will make everyone else do disproportionately more work than you to build upon it. In either case you are the beneficiary.

It is an unfortunate convention of science that research should pretend to be reproducible; our top tips will help you salve the conscience of certain reviewers still bound by this fussy conventionality, enabling them to enthusiastically recommend acceptance of your irreproducible work.

1. **Think “Big Picture”.** People are interested in the science, not the dull experimental setup, so don’t describe it. If necessary, camouflage this absence with brief, high-level details of insignificant aspects of your methodology.

2. **Be abstract.** Pseudo-code is a great way of communicating idiosyncratic





SHARE PERSPECTIVE



Reproducible Research in Computational Science

Roger D. Peng

See all authors and affiliations

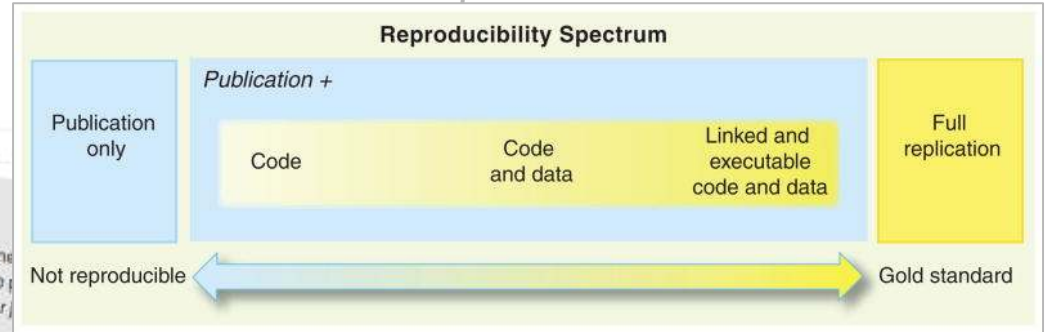
Science 02 Dec 2011:
Vol. 334, Issue 6060, pp. 1226-1227
DOI: 10.1126/science.1213847

Article

Figures & Data

Abstract

Computational science has led to exciting new exposed limitations in our ability to evaluate potential to serve as a minimum standard for replication of a study is not possible.



The rise of computational science has led to exciting and fast-moving developments in many scientific areas. New technologies, increased computing power, and methodological advances have dramatically improved our ability to collect complex high-dimensional data (1, 2). Large data sets have led to scientists doing more computation, as well as researchers in computationally oriented fields directly engaging in more science. The availability of large public databases has allowed for researchers to make meaningful scientific contributions without using the traditional tools of a given field. As an example of this overall trend, the growing availability of publicly available astronomical data has enabled researchers to make discoveries that were previously impossible.

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PERSPECTIVE

Good enough practices in scientific computing

Greg Wilson Jennifer Bryan Karen Cranston Justin Kitzes Lutz Niederbräugl Tracy K. Teal

Published: June 22, 2017 • <https://doi.org/10.1371/journal.pcbi.1005510>

Article	Authors	Metrics	Comments	Related Content
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ADVERTISING

Author summary

Computers are now essential in all branches of science, but most researchers are never taught the equivalent of basic lab skills for research computing. As a result, data can get lost, analyses can take much longer than necessary, and researchers are limited in how effectively they can work with software and data. Computing workflows need to follow the same practices as lab projects and notebooks, with organized data, documented steps, and the project structured for reproducibility, but researchers new to computing often don't know where to start. This paper presents a set of good computing practices that every researcher can adopt, regardless of their current level of computational skill. These practices, which encompass data management, programming, collaborating with colleagues, organizing projects, tracking work, and writing manuscripts, are drawn from a wide variety of published sources from our daily lives and from our work with volunteer organizations that have delivered workshops to over 11,000 people since 2010.

Figures



Reader Comments (1)

Media Coverage (0)

Figures

Subject Areas

- Computer software
- Control systems
- Data management
- Data processing
- Software tools
- Programming language
- Source code
- Reproducibility

Archived Tweets

The Four S's of Real Data – And a Need for Data Technicians, Not Data Scientists?

"Meh" to the 4 Vs of "big data", for most people, most of the time, *real data* is:

- *small*: a few rows and a few columns;
- *slow*: comes out rarely, (period),
- *spreadsheeted*: it just is.
- *smelly*: indications in the smells, spreadsheet smells

At the same time, all data actually got round to using same as "J Smith"?; data-month first? Is it the date of a database, or a well formed easily work with), and so on.

If the value you think you want from, and what you pay your, data scientist for is the stats'n'insights'n'data mining stuff, then should they be spending most of their time doing the grunt work, much of which relies on craft knowledge and skills? How many data scientists do we actually need if they aren't spending all their time poking around fixing the plumbing?

Don't we need more *data technicians* or *data tech eng's* (technical engineers) who can do the labour intensive stuff well (using their craft knowledge) as well as making a bit of sense from it (getting a bit of "insight" out of it based on familiarity with it) using the real data every company has? I just don't get this whole "data science" hype thing... More people need to fix a dripping tap than a leaking high pressure, superheated steam valve in an online nuclear power station. So why the hype about a huge skills gap in the latter when what every company needs is someone who can do the former?

- *small*: a few rows and a few columns;
- *slow*: comes out rarely, often according to a trailing *schedule* (once a week, once a month, some time after the reported period),
- *spreadsheeted*: it just is...
- *smelly*: indications in the data that something is wrong with the way it's been collected, processed or analysed. (Cf. code smells, spreadsheet smells).



Why has it not happened yet? Landmark papers

Electronic Documents Give Reproducible Research a New Meaning

Joe N. Claerbout and Martin Arnsperger, Stanford Univ.

REPL

Statistical Analyses and Reproducible Research

Robert GENTLEMAN and Duncan TEMPLE LANG

Claerbout, J., & Karrenbach, M. (1992).
Electronic documents give reproducible
research a new meaning. SEG Technical
Program Expanded Abstracts, 601–604.

<https://doi.org/10.1190/1.1822162>

Gentleman, R., & Temple Lang, D. (2007).
Statistical analyses and reproducible
research. Journal of Computational and
Graphical Statistics, 16(1), 1–23.

<https://doi.org/10.1198/106186007X178663>

SUMMARY

A revolution in education and technology has led to the free exchange of digital information and access to online data. In the exchange of information, the user is no longer limited to a single source of information. The user is now the information, the data, the process, and the program. The user is now the information, the data, the process, and the program. The user is now the information, the data, the process, and the program.

In 1992, we began developing a CD-ROM. The CD-ROM was a collection of documents, data, and programs. The CD-ROM was a collection of documents, data, and programs. The CD-ROM was a collection of documents, data, and programs.

In 1992, we began developing a CD-ROM.

• Users have to manage a collection of data, which is not only a collection of data, but also a collection of data.

• To be reproducible, the user must be able to access the data, which is not only a collection of data, but also a collection of data.

• Users have to be able to access the data, which is not only a collection of data, but also a collection of data.

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It is important, if not essential, to integrate the computations and code used in data analyses, methodological descriptions, simulations, and so on with the documents that describe and rely on them. This integration allows readers to both verify and adapt the claims in the documents. Authors can easily reproduce the results in the form, and they can present the document's contents in a different medium, for example, with interactive content. This article describes a new way to write documents for both authoring and distributing these integrated, dynamic documents that contain text, code, data, and any auxiliary content needed to recreate the computations. The documents are dynamic in that the contents—including figures, tables, and so on—can be reconstructed each time a view of the document is generated. One model treats a dynamic document as a master or "source" document from which one can generate different views in the form of individual, derived documents for different audiences.

We introduce the concept of a *compendium* as a container for one or more dynamic documents and the different elements needed when processing them, such as code and data. The compendium serves as a means for distributing, reorganizing, and updating the collection.

The step from disseminating analyses via a compendium to reproducing it is needed to conduct fully reproducible research, so more modern papers, with accompanying software tools that allow the reader to directly reproduce the results and employ the computational methods that are presented in the research paper. Some of the issues involved in preparing for the production, distribution, and use of such reproducible research are discussed.

Key Words: Compendium; Dynamic documents; Dynamic programming; Modeling; Programming; Perl; Python; R.

1. INTRODUCTION

Statistical methodology generally involves algorithmic concepts. The descriptions of how to perform a specific analysis often are given in detail or, generally, how to perform a type of analysis tend to be similarly procedural or algorithmic. Expressing these concepts in a

Robert Gentleman is Professor, Program in Computational Biology, Division of Public Health Sciences, Fred Hutchinson Cancer Research Center, 1100 Fairview Avenue North, M2-B105, PO Box 357234, Seattle, WA 98105-7234 (e-mail: rgentleman@fredhutch.org); Duncan Temple Lang is Assistant Professor, Department of Statistics, University of California, Davis, Box 98016, Davis, CA 95616 (e-mail: dtemplelang@ucdavis.edu).

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Journal of Computational and Graphical Statistics, Volume 16, Number 1, Pages 1–23
DOI: 10.1198/106186007X178663

File names



http://www2.stat.duke.edu/~rcs46/lectures_2015/01-markdown-git/slides/naming-slides/naming-slides.pdf



Dan Lurie
@danjlurie

Folgen

Well-organized directory structures + informative file names can make a big difference in productivity and reproducibility.

Jenny Bryan @JennyBryan

Low-tech common sense about filenames. My slides may be homely, but this practical advice holds up well. www2.stat.duke.edu/~rcs46/lecture ...



Dan Lurie @danjlurie · 4 Std.

I recently came back to a project from a few months ago and I am so proud of past Dan.

Original (Englisch) übersetzen



Jenny Bryan
@JennyBryan

Folge ich

Low-tech common sense about filenames. My slides may be homely, but this practical advice holds up well. www2.stat.duke.edu/~rcs46/lecture ...

Original (Englisch) übersetzen

19:32 - 17. Aug. 2017

94 Retweets 239 „Gefällt mir“-Angaben

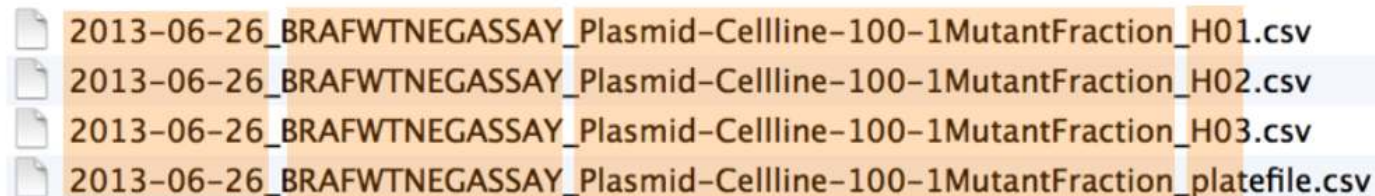


14 94 239

<https://twitter.com/JennyBryan/status/898372123281858560>

Jenny Bryan's principles for file names

machine readable (regex + globbing for search & filter & extract. CLI & R. vs. -)



2013-06-26_BRAFWTNEGASSAY_Plasmid-Cellline-100-1MutantFraction_H01.csv
2013-06-26_BRAFWTNEGASSAY_Plasmid-Cellline-100-1MutantFraction_H02.csv
2013-06-26_BRAFWTNEGASSAY_Plasmid-Cellline-100-1MutantFraction_H03.csv
2013-06-26_BRAFWTNEGASSAY_Plasmid-Cellline-100-1MutantFraction_platefile.csv

human readable (embrace the slug)

plays well with default ordering

(ISO8601, left pad numbers)

*Easy to implement NOW,
pays off later!*

```
01_marshall-data.r  
02_pre-dea-filtering.r  
03_dea-with-limma-voom.r  
04_explore-dea-results.r  
90_limma-model-term-name-fiasco.r  
helper01_load-counts.r  
helper02_load-exp-des.r  
helper03_load-focus-statinf.r  
helper04_extract-and-tidy.r
```